

THE **AI-ENABLED** ENTERPRISE

From Generative AI to Intelligent Agents
and Autonomous Operations

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Introduction

Artificial intelligence is changing quickly. What began for many businesses as a way to generate text, summarize information, or answer questions is becoming something much more significant.

AI can now help people interact with information and technology using everyday language, reducing some of the technical barriers that traditionally separated business needs from the systems used to address them. This idea was at the heart of Generative Integration, using generative AI and large language models to make it easier to connect systems, data, and business processes.

Today, AI is moving beyond generating information toward helping businesses get work done. It can increasingly interact with business systems, assist with decisions, coordinate activities, and take actions within defined boundaries. This evolution is giving rise to AI agents, intelligent automation, multi-agent systems, and new possibilities for more autonomous operations.

For business leaders, the opportunity is not simply to adopt another technology. It is to look differently at how the organization works: how customers are served; how employees find information; how decisions are made; how problems and exceptions are handled; and where time, effort, and resources could be used more effectively.

Throughout this book, you will discover how artificial intelligence is evolving, what it can realistically do for your organization, and how it can be applied to solve real business problems and create measurable value. Through plain language explanations, practical examples, and real-world business scenarios, we will move from understanding AI to recognizing opportunities and considering how they can be put into practice.

The journey begins with understanding what has changed and why it matters for business.

UNDERSTANDING THE AI TRANSFORMATION

From Generative AI to the AI-Enabled Enterprise

Artificial intelligence has moved rapidly from a specialized technology to something millions of people can use through everyday language. A person can ask a question, summarize a document, analyze information, generate an image, write software, or explore an idea simply by describing what they want. For businesses, however, the more important development is not simply that AI can generate better content. AI is increasingly able to connect with business information, interact with enterprise systems, assist with decisions, use software tools, and take actions. This expands the opportunity from using AI to generate information to using AI to help accomplish work. Together, these capabilities point toward a broader transformation in how people, technology, information, and business processes can work together.

What Is Generative AI?

Generative AI refers to artificial intelligence systems capable of creating new content or responses based on information, instructions, and patterns learned from large amounts of data. Depending on the system, this can include text, images, software code, audio, video, analysis, and other forms of content. Large language models have made these capabilities particularly accessible because people can interact with them using natural language. Instead of learning specialized commands or navigating complex software interfaces, a person can describe what they need, provide relevant information, ask questions, refine the response, and continue the interaction conversationally.

In a business environment, this can help people create content, summarize documents, analyze information, retrieve knowledge, and prepare reports. These capabilities can save time and make information easier to use, but generating information represents only one part of the opportunity. The larger change begins when AI becomes connected to the information, systems, and processes through which organizations actually operate.

From Generative AI to Generative Integration

Businesses rarely operate through a single system. Customer, financial, operational, and organizational knowledge is often distributed across different applications and information sources. Employees frequently become the connection between them, searching for information, transferring data, interpreting what they find, and determining what should happen next. Generative Integration extends generative AI into this environment by using AI to make it easier to connect information, systems, and business processes. Instead of only generating a response, AI can help interpret what a person is trying to accomplish, retrieve relevant information, transform data, interact with applications, and support the execution of work.

Consider a manager who asks:

“Show me the customers with unresolved service issues who also have important opportunities in the sales pipeline.”

Answering that question may require information from customer service and CRM systems, matching records, interpreting their status, and presenting the result in a useful form. The person does not necessarily need to know how those systems store their data or how the integration works. Natural language becomes a way to express the business need while AI helps bridge the gap between that need and the underlying technology. Integration therefore becomes more than a technical concern; it becomes part of how people access information and coordinate work across the enterprise.

From Generating Information to Taking Action

Connecting AI to enterprise information creates another possibility, AI can help do something with what it understands. A system might retrieve a customer record, determine that follow-up is required, prepare a response, schedule an appointment, update the CRM, or initiate another business action.

This represents an important shift:



Traditional automation can already execute many business activities, but it generally follows predefined rules and workflows. AI adds the ability to interpret less structured information, consider context, evaluate alternatives, and assist with situations that cannot always be described through rigid rules alone. The distinction becomes especially important when the work contains decisions and exceptions. A standard workflow may process an order when everything proceeds normally, but when inventory is unavailable, a supplier is delayed, the customer has an unusual requirement, or several alternatives are possible, someone may need to investigate the situation and decide what should happen. AI can increasingly assist with this kind of work.

From AI Agents to Autonomous Operations

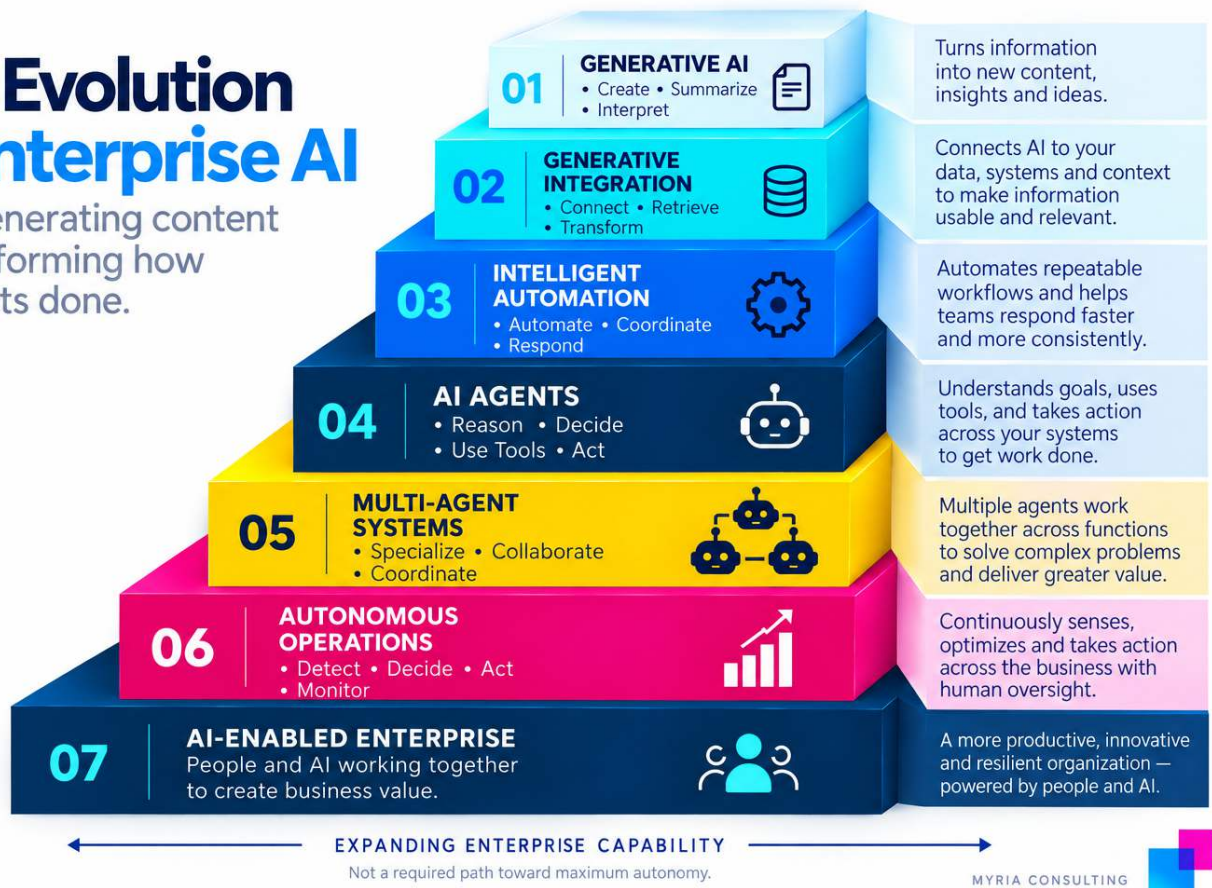
An AI agent extends these capabilities by working toward an objective rather than simply producing a response. It can gather information, reason about a situation, use approved tools, take authorized actions, observe the result, and determine what should happen next within defined boundaries. For example, an agent responsible for sales leads could review a new prospect, conduct follow-up, determine whether the opportunity qualifies, schedule an appointment, update the CRM, and involve a salesperson when necessary. The objective is not to generate a message, but to move the opportunity toward an appropriate outcome.

As work becomes more complex, specialized agents can coordinate. One agent might focus on customer communication, another on inventory, another on supplier information, and another on fulfillment. Together, they can contribute to a broader business objective while people remain involved where judgment, authority, expertise, or accountability is required. When these capabilities are applied continuously to operational decisions and exceptions, organizations begin moving toward autonomous operations: environments in which AI can help detect what requires attention; understand the situation; evaluate alternatives; take authorized action; monitor the result; and involve people when necessary.

FIGURE 1.1

The Evolution of Enterprise AI

From generating content to transforming how work gets done.



This evolution represents expanding capability, not a requirement that every organization or process move through every stage. Different business needs will require different combinations of AI, automation, and human involvement.

What Is an AI-Enabled Enterprise?

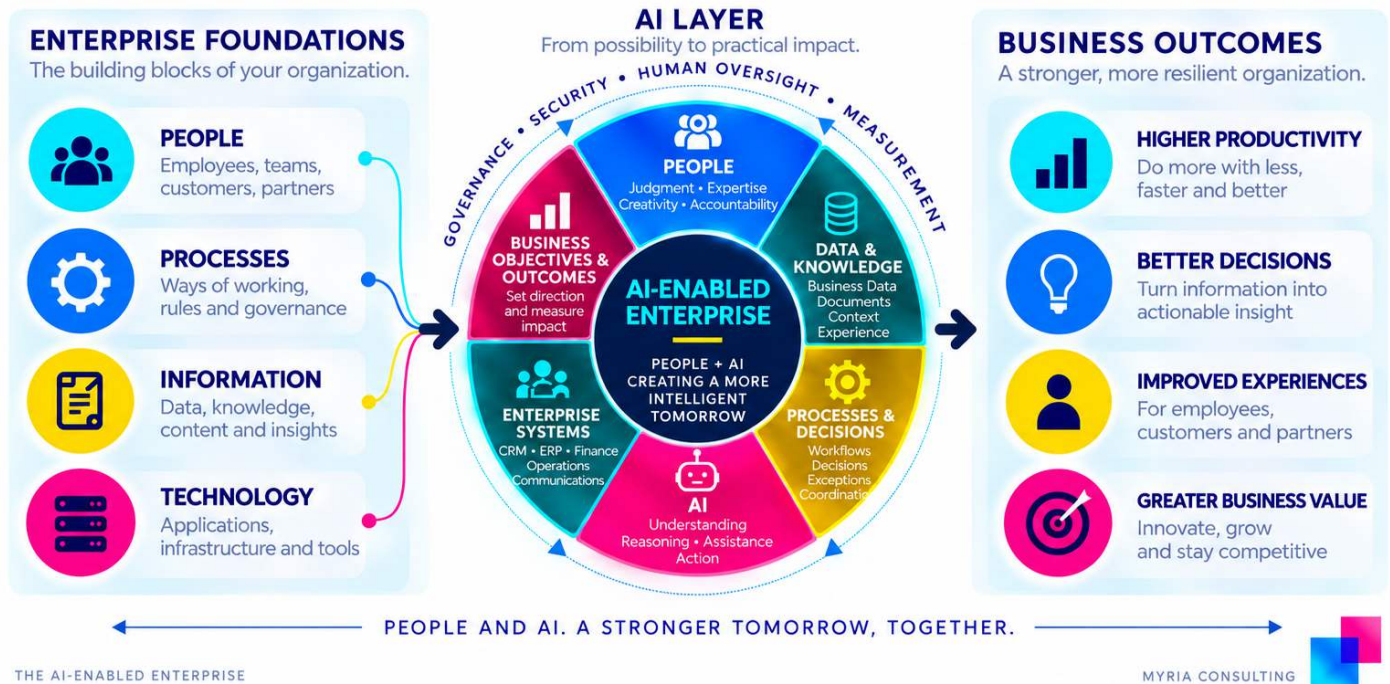
An AI-enabled enterprise does not simply use AI tools. Instead, it integrates people, processes, data, knowledge, and systems with artificial intelligence to optimize workflows and elevate business outcomes. AI may help employees find information, analyze situations, prepare recommendations, automate repetitive cognitive work, coordinate activities, interact with systems, or manage certain decisions and exceptions within established boundaries, while people continue to provide judgment, expertise, creativity, relationships, accountability, and oversight where those capabilities matter most. The important question is therefore not how much AI an organization can deploy, but where AI can improve the way the organization operates and creates value.

FIGURE 1.2

The AI-Enabled Enterprise

Connecting people, processes, information and technology to create business value.

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An AI-enabled enterprise isn't built by technology alone. True business value is unlocked when AI orchestrates people, workflows, data, and systems.

What This Means for Business Leaders

The question is no longer limited to what generative AI can produce. Leaders can examine where information is difficult to access, decisions are delayed, systems remain disconnected, repetitive work consumes capacity, exceptions require frequent intervention, or customers and employees experience unnecessary friction. Some of these opportunities may require better processes rather than AI; others may be addressed through conventional automation, system integration, generative AI, AI assistance, or agents capable of taking action. **The objective is not to introduce the most advanced technology possible, but to select the capability appropriate to the business problem and the outcome being pursued.**

Becoming AI-enabled is therefore not simply a technology initiative. It involves understanding work, improving access to information, connecting systems, determining how decisions should be made, defining appropriate roles for people and AI, establishing governance, measuring results, and developing the organization's ability to learn from experience. **The evolution from generative AI toward more intelligent and autonomous capabilities creates an opportunity to reconsider not only what AI can generate, but what people and organizations can accomplish with it.**

The starting question is therefore simple:

Where could AI help us work better, make better decisions, serve customers better, or create greater value?

THE FOUNDATIONS OF INTELLIGENT BUSINESS

Natural Language as the New Business Interface

For most of the history of business technology, people have had to learn how software works in order to use it. They navigate menus, complete forms, construct searches, and move between applications to find information or complete a task. Artificial intelligence introduces a new possibility: people can express what they need in everyday language; AI then interprets the request and translates it into information, analysis, or action. Natural language can become an interface between a person's business intent and the technology capable of fulfilling it.

From Commands to Intent

Traditional software requires users to express their needs in ways the system understands. A manager looking for a particular group of customers may need to know which application contains the information; which fields to select; and how to construct the report. With AI, the manager may instead ask:

“Which customers have reduced their purchases over the last six months and currently have unresolved service issues?”

Users express business intent, not the technical steps required to find an answer. AI bridges this gap by interpreting the request, retrieving the right data, running the analysis, and formatting the output. This is vital because instead of navigating database fields or software code, human leaders solve problems, mitigate risks, and chase outcomes. By communicating through natural language, employees leverage technology using everyday business vocabulary, unlocking absolute autonomy independent of IT departments.

Understanding Everyday Business Language

Human language is flexible and often incomplete. People use abbreviations, context, industry terminology, and expressions whose meaning depends on the situation. A request such as “Show me the accounts we should be worried about” may require considering customer activity, service issues, payment history, sales information, or other context. Large language models can interpret this language, identify what a person is trying to accomplish, recognize relevant concepts, ask for clarification when necessary, and translate the request into a form that other systems or tools can use. AI will not always interpret a request correctly, particularly when it is ambiguous or consequential, but the interaction can begin with what the person is trying to accomplish rather than with the technical structure of the system.

From Conversation to Business Capability

Natural language becomes more powerful when AI is connected to business knowledge, data, and systems. Through the same conversational interaction, a person can ask about a customer, retrieve a policy, analyze data, prepare a report, schedule an appointment, update a record, and initiate other business activities.

Consider an employee who asks:

“A customer says their order is late. Find out what happened and tell me what we can do.”

AI may need to identify the customer and order, retrieve operational information, determine what happened, consider available alternatives, and present an appropriate recommendation. If authorized, an AI agent may also take action or involve the appropriate person. The conversation is therefore only the visible interface. Behind it may be organizational knowledge, data, enterprise systems, automation, AI agents, and human involvement. The true value of natural language is not simply that people can talk to AI, but that conversation becomes a powerful interface for accessing organizational capabilities.

FIGURE 2.1

Natural Language as the Business Interface

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Making Technology More Accessible

Natural-language interaction can reduce some of the technical barriers separating people from information and software capabilities. Employees may explore information without constructing complex queries, managers may ask questions without requesting new reports, and subject-matter experts may interact with systems without learning every technical detail behind them. Business users can also describe requirements, explore possibilities, review results, and refine them through conversation, allowing people who understand the work to participate more directly in designing how technology supports it. Technology specialists remain essential for architecture, integration, security, data quality, governance, reliability, and complex implementation. Natural language does not eliminate the need for technical expertise; it can reduce the technical effort required for people to express and explore business needs.

A New Interface for Work

Natural language will not replace every business interface. Forms, dashboards, specialized applications, and structured workflows remain valuable when they provide faster, clearer, or more controlled ways to perform a task. The opportunity is to use conversation where it makes interaction with information and systems easier and more effective.

The significance of natural language therefore extends beyond better chatbots. It allows people to begin with **what they want to know or accomplish** rather than how the underlying software works. As AI becomes connected to knowledge, data, tools, and enterprise systems, conversation can move from answering questions to helping people perform work.

Natural language is becoming a new business interface: a way for people to express intent, access organizational capabilities, and directly translate goals into information, decisions, and action.

Connecting AI to the Enterprise

AI becomes significantly more useful when it can work with the information and systems a business already uses. A standalone AI system may answer questions, generate content, or analyze information provided to it, but many valuable business activities require access to customer records, orders, inventory, financial information, policies, documents, schedules, communications, and operational systems. Connecting AI to the enterprise allows it to move from general intelligence toward useful business capability.

Connecting AI to Business Systems

Most organizations operate through many applications. Customer relationships may be managed in a CRM, orders and inventory in an ERP or operational system, financial information in accounting platforms, documents in knowledge repositories, and communication through email, messaging, and collaboration tools. Employees routinely move between these systems to gather information, reconcile differences, update records, and complete work.

AI can provide a more intelligent layer across this environment. Instead of requiring a person to search several applications, AI can identify what information is needed, retrieve it from appropriate sources, combine it, interpret the situation, and present the result. When connected to approved tools and APIs, it can also initiate actions in those systems.

Consider a salesperson who asks:

“What is happening with this customer before I call them?”

A useful response may require more than information from the CRM. AI could bring together recent purchases, open opportunities, service issues, outstanding invoices, previous communications, and other relevant information to provide a concise picture of the relationship. The value comes not from replacing the underlying systems, but from **making their combined information easier to access and use.**

Connecting Data, Knowledge and Action

Enterprise information exists in different forms. Some is structured, such as customer records, transactions, inventory levels, prices, and financial data. Other information is contained in documents, policies, contracts, procedures, emails, and reports. Many business decisions require both data and context. Knowing that an order is delayed may come from an operational system, while understanding what can be done about it may require a service policy, customer commitment, supplier information, or previous communication. Connecting these sources gives AI the context needed to provide more useful assistance.

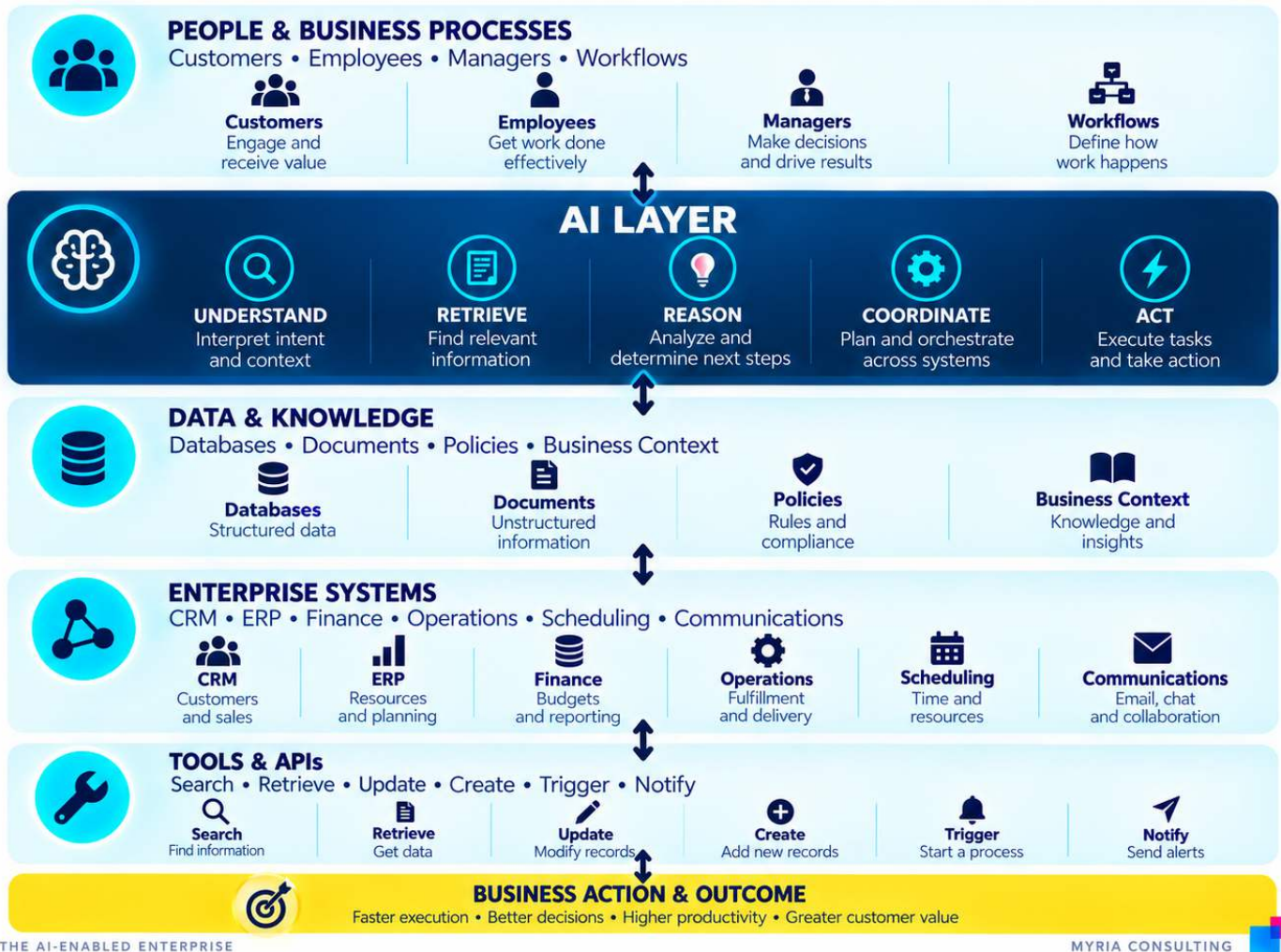
The next step is connecting understanding to action. If AI determines that a customer requires follow-up, it may prepare a response, create a task, update a record, schedule an appointment, or initiate an approved workflow. The enterprise connection therefore has three dimensions: what AI can know, what it can understand, and what it is allowed to do.

FIGURE 3.1

The AI Integration Layer

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Connecting people, data, systems and tools to turn intelligence into real business outcomes.



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The AI layer does not replace enterprise applications. It connects human intent, organizational information, and system capabilities so they can be used together more effectively.

APIs, Tools and Enterprise Actions

For AI to interact with business systems, it needs controlled ways to access their capabilities. One of the most common is an **application programming interface, or API**, which allows software systems to exchange information or request actions from one another. Rather than giving AI unrestricted access to an application, organizations can provide specific tools that perform approved functions through these interfaces.

A customer-service agent might have tools to retrieve a customer, check an order, create a service case, or send a notification, while a sales agent might find a prospect, update a lead, schedule a follow-up, or book an appointment. Organizations can define the available actions, permissions, validation rules, and situations requiring human approval. AI should receive the capabilities required for the work, with access and authority appropriate to the task.

Working Across the Enterprise

Once AI can work across systems, data, knowledge, and tools, it can help coordinate activities that cross organizational and technological boundaries. A delayed customer order, for example, may require checking the order, inventory, supplier information, shipping activity, customer commitments, and service policies. Instead of an employee manually assembling this information, a connected AI capability can help gather the relevant context, identify the issue, evaluate available options, and support the appropriate response.

The underlying systems remain specialized, but **AI can connect their information and capabilities around the business problem being solved**. Enterprise integration therefore becomes more than moving data between applications; it can help the organization understand situations and coordinate what should happen next.

Design for Reliability and Control

Connecting AI to enterprise systems also requires appropriate controls. Organizations need to determine which information AI can access, which systems it can interact with, what actions it can perform, and when human involvement is required. Identity, permissions, security, privacy, data quality, monitoring, and auditability therefore become part of the design.

Broader access does not automatically create better AI. A system answering questions about company policies may need only approved documents, while an agent managing customer appointments also requires CRM and scheduling capabilities. **The objective is to provide the information and capabilities required for a particular business purpose while maintaining appropriate control.**

Connecting AI to the enterprise therefore means creating a controlled connection between **people, information, systems, and action**. Once that connection exists, AI can move beyond answering isolated questions and begin helping people coordinate work across the organization. **The real value of enterprise AI emerges when it can access the right information; understand the business context; and connect that understanding to the systems and actions required to produce an outcome.**

From Workflows to Intelligent Automation

Businesses have used automation for decades to perform repetitive work faster, more consistently, and at greater scale. Rules, scripts, workflow platforms, and enterprise systems can move information, update records, route requests, generate notifications, and execute predefined processes with little human intervention. These capabilities remain valuable, but AI expands what can be automated by helping systems work with language, context, decisions, and exceptions that traditional automation often struggles to handle.

Traditional Business Automation

Traditional automation works particularly well when a process is predictable and its rules can be defined in advance. An organization might automatically send an invoice when an order is completed, route a purchase request according to its value, or update inventory after a shipment.

The system follows a predetermined sequence:



This approach can produce significant gains in speed, consistency, cost, and reliability, but it becomes more difficult when a situation does not fit the expected path. Missing information, unusual customer requests, supplier delays, or other exceptions may require someone to investigate what happened and decide what to do. **Traditional automation is strongest when the organization already knows the steps; AI becomes particularly useful when determining the appropriate next step requires interpretation or judgment.**

From Fixed Workflows to Intelligent Workflows

AI can extend traditional workflows by handling parts of the process that require understanding less structured information or considering context. Instead of relying exclusively on predefined conditions, an AI-enabled workflow can interpret a request, classify information, identify relevant knowledge, evaluate alternatives, recommend an action, or determine which predefined path should be followed.

Consider a customer requesting a refund. Traditional automation may check whether the request falls within a specified period and automatically approve or reject it. An AI-enabled process could also consider the customer's explanation, transaction history, relevant policy, and previous interactions before recommending what should happen. The workflow still provides structure and control, while AI helps with the parts requiring interpretation.

The distinction can be summarized as:



This combination can make automation useful for a broader range of work without requiring AI to control the entire process.

Managing Exceptions and Human Decisions

Many business processes work efficiently until something unexpected happens. An order cannot be fulfilled, a payment does not match an invoice, a supplier misses a commitment, information is incomplete, or a customer requests an exception. These situations often interrupt automated workflows and require people to gather information and determine how to proceed.

AI can help by detecting the exception, gathering relevant information, interpreting what happened, identifying possible responses, and recommending or preparing the next action. Depending on the organization's rules and the consequences of the decision, AI may execute an approved action or escalate the situation to a person. If an order cannot be fulfilled from its original location, for example, AI might check inventory elsewhere, review delivery commitments, evaluate alternatives, and recommend a response. A routine, low-risk decision might be handled automatically, while one involving significant cost, unusual circumstances, or an important customer could be referred to an employee.

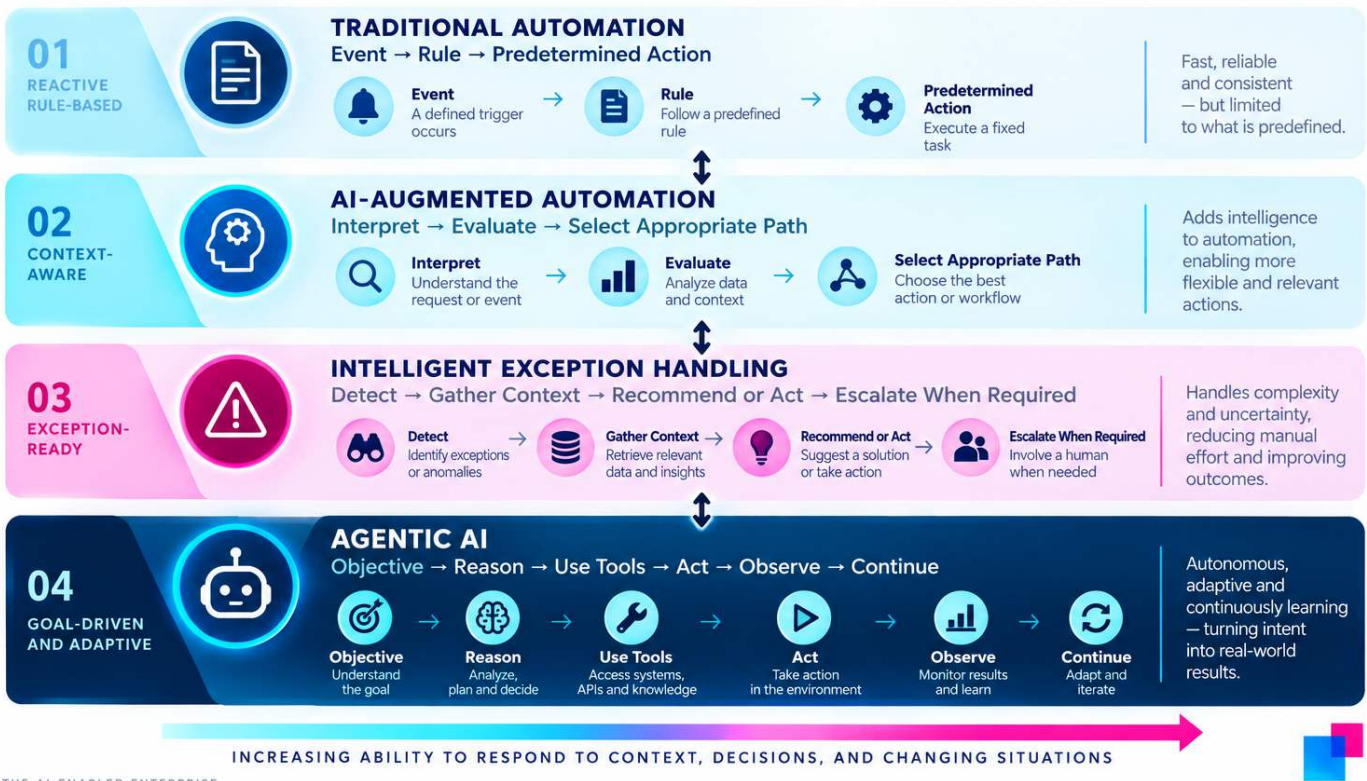
The objective is not to remove people from every exception, but to reduce the effort required to understand and resolve them while directing human attention toward situations where judgment, authority, expertise, or relationships matter most.

FIGURE 4.1

From Traditional Automation to Agentic AI

A progression from rule-based execution to adaptive, goal-driven action.

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This progression does not make traditional automation obsolete. **Each approach is appropriate for different kinds of work, and they can operate together within the same business process.**

From Automation to Agentic AI

AI-augmented automation generally operates within a workflow designed in advance: the process defines the overall path while AI provides intelligence at particular points. **Agentic AI** goes further by giving an AI agent an objective and allowing it to determine, within defined boundaries, which steps and tools are required to pursue it.

A traditional sales workflow, for example, might send a follow-up message three days after a new lead arrives. An agent responsible for helping qualified prospects reach the appropriate next step could instead review the lead, conduct follow-up, determine qualification, schedule an appointment, update the CRM, and involve a salesperson when necessary. The sequence can change according to what happens.

The agent still operates within defined objectives, information access, tools, permissions, approval requirements, and escalation conditions. What changes is that the organization does not need to prescribe every step in advance. **The agent can determine what should happen next within the boundaries established by the organization.**

Traditional automation, AI-augmented workflows, and agentic AI therefore complement one another. Predictable, rules-based activities may remain best suited to conventional automation; AI can strengthen workflows requiring interpretation and exception handling; and agents can address work where achieving an objective requires a more dynamic sequence of decisions and actions.

Traditional automation asks, **“What steps should the system execute?”** Agentic AI introduces a broader question: **“What needs to be accomplished, and what should happen next?”** This ability to pursue an objective, gather information, reason about a situation, use tools, take actions, and respond to results is the foundation of the AI agent.

THE AGENTIC ENTERPRISE

Understanding AI Agents

Artificial intelligence becomes more capable when it can move beyond responding to individual requests and begin working toward an objective. **An AI agent is an AI system that can work toward a defined objective by gathering information, determining what to do, using approved tools to act, and adjusting based on the results.**

Consider a new sales lead arriving through a company's website. An AI assistant could summarize the lead for a salesperson. An AI agent could go further by checking whether the prospect already exists in the CRM, asking qualification questions, evaluating responses, updating the customer record, scheduling meetings, and involving a salesperson when necessary. The agent is not simply completing one task; it is coordinating several activities toward an objective.

From AI Assistants to AI Agents

The terms assistant, copilot, and agent are used somewhat differently across the AI industry, but they provide a useful way to understand different levels of AI participation in work. An assistant helps you do the work by responding to questions and instructions. A copilot works with you as you do the work, providing information, analysis, recommendations, or content while you remain in control. An agent can perform parts of the work on your behalf, determining what needs to happen and taking authorized actions within defined boundaries.

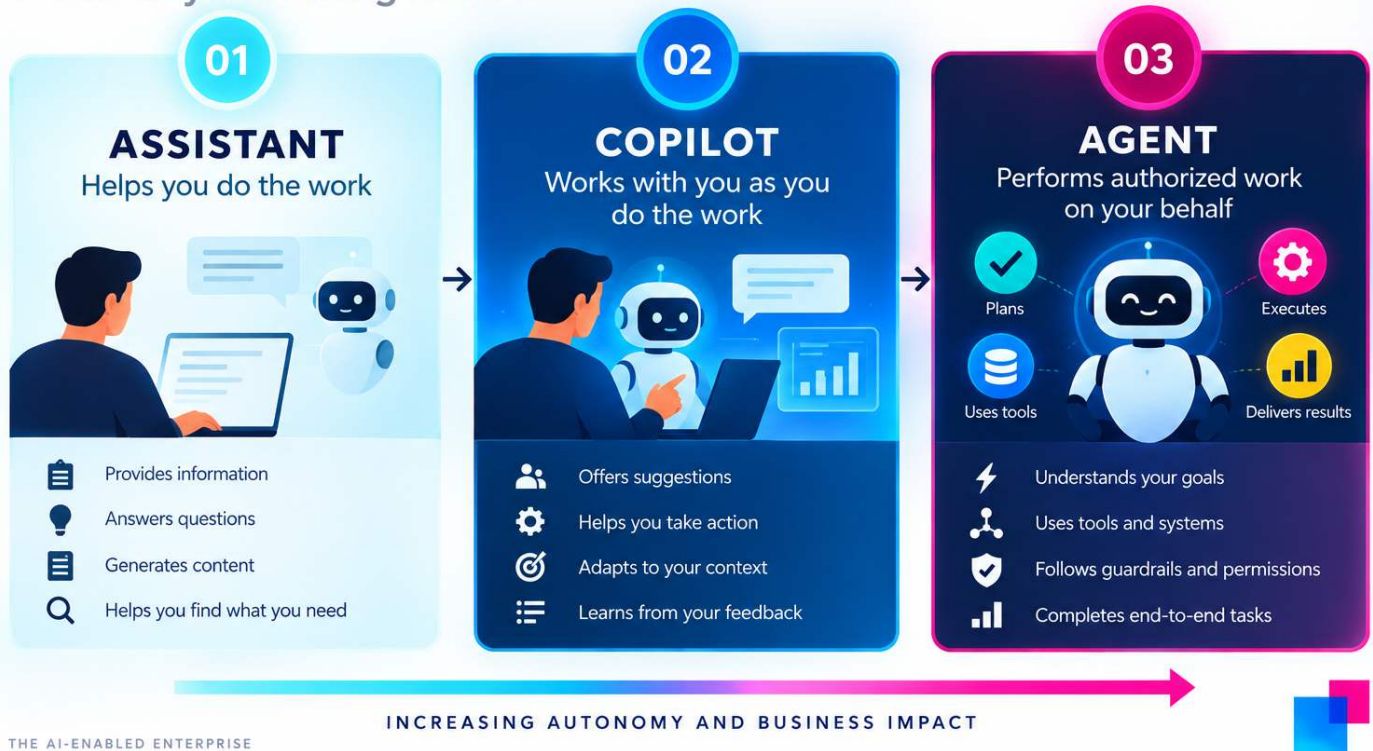
An assistant might tell a salesperson which customers require follow-up. A copilot might prioritize those customers and prepare suggested communications. An agent could identify the customers, determine the appropriate follow-up, use approved tools to initiate actions, monitor the results, and involve the salesperson when necessary. The important difference is how much responsibility AI has for moving the work toward an outcome.

FIGURE 5.1

From Assistant to Agent

A new way of working with AI.

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An agent is not automatically better than an assistant, copilot, or traditional automation. The right approach depends on the work. Agents become particularly useful when achieving an objective requires several steps, information from different sources, decisions along the way, actions across systems, and the ability to adapt as the situation develops.

How an AI Agent Works

An agent begins with a goal: the outcome it is expected to achieve or help achieve. The goal might be to resolve a customer request, qualify a sales opportunity, investigate an operational problem, or prepare an account for renewal. The agent uses its instructions and available information to determine what needs to happen next. AI reasoning in this context does not mean human understanding or judgment. It refers to using available information, instructions, rules, and AI capabilities to determine an appropriate course of action.

A useful way to understand an AI agent is to think about what a new employee needs before performing a job. A new employee needs to understand what they are expected to accomplish, receive instructions, have access to the information and systems required for the work, know what they are authorized to do, and understand when a decision should be referred to someone else. An AI agent requires a similar operating structure: a goal, instructions, relevant context and knowledge, access to approved tools, clear boundaries, and a way to escalate situations it should not handle independently. The comparison has limits, however. An AI agent does not bring human judgment, accountability, experience, or understanding to the work. These differences are why its responsibilities, access, controls, and human oversight must be deliberately designed.

To perform useful work, an agent may need tools that provide controlled access to business capabilities. A tool might allow the agent to retrieve a customer record, check inventory, search enterprise knowledge, create a service request, update a CRM record, or schedule an appointment. The agent can select an appropriate tool, use it, examine the result, and determine what should happen next.

The agent also needs context: the information necessary to understand the current situation. This might include the user's request, customer information, relevant policies, previous actions, and information retrieved from business systems. **Memory allows appropriate information to remain available beyond the immediate interaction.** Context might contain what the agent needs to resolve a customer's current request, while memory could retain approved information from previous interactions that may be useful later.

Some objectives also require planning. For a simple task, the agent may only need to determine the next appropriate action and then respond to the result. More complex tasks may require several coordinated steps. The agent can gather information, evaluate what it finds, decide what to do, take an action, and then determine whether additional action is required.

These capabilities form a practical operating cycle:

FIGURE 5.2

Anatomy of an AI Agent

From goal to action — a continuous cycle of reasoning, action and learning.

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The agent then evaluates the result. Did the action succeed? Has the objective been achieved? Is another action required? Does the situation require human involvement? The result becomes new information that can influence what happens next.

A simple model for understanding this behavior

FIGURE 5.3

The AI Agent Operating Loop

A continuous cycle that turns intent into real-world outcomes.

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This ability to determine the next appropriate action based on what happens is one of the important differences between an agent and a fixed workflow.

For example, an agent helping resolve an order problem might retrieve the order, discover that an item is unavailable, check inventory elsewhere, review incoming deliveries, and identify alternatives. If a routine inventory transfer is permitted within established rules, the agent might initiate it. If the solution would create significant additional cost or affect an important customer commitment, it could instead recommend an action and request employee approval. The objective remains the same, but what happens next depends on the situation.

Boundaries and Human Oversight

Giving AI the ability to act makes **boundaries and human oversight essential parts of agent design**. An agent should have a defined purpose, access only to the information and tools it needs, and clear limits on the decisions and actions it can perform. A sales agent might be permitted to create leads, update routine CRM information, and schedule appointments but not change prices or approve contractual terms. A customer-service agent might resolve common requests but require approval before issuing a significant refund.

Human involvement can vary according to the importance, uncertainty, and risk of the decision. AI may provide information, recommend an action for approval, execute routine actions within established limits, or escalate situations requiring human judgment. People establish the objectives and boundaries, retain responsibility for consequential decisions, handle exceptions, and monitor how agents perform.

An effective agent must also know when to stop. It should recognize when the objective has been achieved, when necessary information is unavailable, when an action repeatedly fails, or when the situation exceeds its authority. **Escalating to a person when human judgment is required is part of performing the work correctly.**

The goal is therefore not maximum autonomy. **It is the right combination of AI capability and human involvement for the work being performed.**

When an AI Agent Makes Sense

Not every process needs an AI agent. If a task can be handled reliably with a simple rule, conventional software, or a fixed workflow, those approaches may be simpler and more appropriate. An agent becomes particularly useful when the work repeatedly requires a combination of **interpretation, information gathering, decision making, system actions, and adaptation.**

The business value should determine whether an agent is worth implementing. Organizations can consider how frequently the work occurs, how much employee time it requires, how long it takes, how much variation it contains, whether the necessary information is available, what actions can safely be delegated, and what measurable improvement could be achieved. Activities involving significant financial, legal, safety, privacy, or reputational consequences may still benefit from AI, but generally require stronger controls and greater human involvement.

The question is therefore not, **“Where can we deploy AI agents?”** It is:

“Where does work require enough interpretation, decision making, coordination, and adaptation that an AI agent could help us achieve the outcome more effectively?”

Used appropriately, agents can reduce the time people spend searching for information, coordinating routine activities, monitoring processes, following up, and resolving predictable situations. This allows employees to concentrate more of their attention on relationships, complex problems, creativity, improvement, and decisions where experience and accountability matter most.

The true value of an AI agent lies not in its autonomy, but in the scale of what people and organizations can achieve through its capabilities.

For an agent to perform meaningful work inside an organization, however, these capabilities must be implemented securely and reliably. The next chapter examines how the models, information, memory, tools, controls, and enterprise systems come together to build AI agents that can act.

Building AI Agents That Can Act

Understanding what an AI agent can do is different from making one reliable enough to perform meaningful work inside an organization. An enterprise AI agent needs more than an AI model. It needs **intelligence, instructions, information, tools, and controls** working together. It must understand what it is trying to accomplish, have access to the information required for the task, interact with approved business systems, operate within clearly defined authority, and handle situations when something does not go as expected. This is what turns an AI capability into an enterprise capability. The objective is not to grant AI unrestricted access to the enterprise, but to provide the specific data and tools it needs to perform useful work within a controlled environment.

Building the Enterprise Agent

At the center of an agent is an AI model capable of interpreting information and determining an appropriate response or next action. Different models have different capabilities, speeds, and costs. A routine classification or information-extraction task may require relatively little reasoning, while investigating an unusual customer problem or evaluating several possible actions may require greater capability. The objective is to use a model appropriate to the work rather than automatically using the most powerful model available. **The model operates according to system instructions that define the agent's role, objectives, responsibilities, and expected behavior.** A customer-service agent, for example, might be instructed to resolve routine requests using approved information, avoid making contractual commitments, request approval for refunds above a specified amount, and escalate situations it cannot resolve.

The agent also needs the right context. Providing everything the organization knows would be unnecessary and could make it harder to identify what matters. Context management means making relevant information available when it is needed: the customer's request, transaction details, applicable policies, previous actions, results returned by tools, and other information required to determine what should happen next. Memory becomes important when information needs to persist. Short-term memory can help an agent keep track of what has happened during a task, while longer-term memory can make approved information from previous interactions available when relevant. Organizations therefore need to determine what should be retained, for how long, where it is stored, and who can access it.

Much of what an agent needs to know will not be contained in the AI model itself. Organizations have policies, procedures, product information, contracts, manuals, reports, and other knowledge that changes over time and is specific to the business. **Knowledge retrieval** allows the agent to find relevant information from approved sources when it is needed. A customer might ask whether a purchase can be returned. The agent could retrieve the current return policy and combine it with the customer's transaction information to determine whether the request meets the applicable conditions. The policy represents enterprise knowledge; the transaction represents current operational information. Useful enterprise agents often need both.

Giving the Agent the Ability to Act

Understanding a situation is only part of the job. To perform work, **an agent needs tools** that provide specific business capabilities, such as finding a customer, retrieving an order, checking inventory, creating a service request, updating a CRM record, scheduling an appointment, or initiating an approved communication. These tools often connect to enterprise applications through **APIs**, which provide controlled ways for systems to exchange information and perform actions. When the agent determines which tool it needs and invokes it, **it is using tool calling to move from reasoning to action.**

The important principle is that an agent should not receive unrestricted access to a CRM, ERP, financial application, or other business system. It should receive only the capabilities required for its role. A sales agent might be able to find a customer, create a lead, update routine information, and schedule a follow-up, while changing prices or approving contractual terms remains outside its authority. Information exchanged between the agent and these tools should also be predictable.

Structured outputs allow systems to require information in a defined format rather than accepting unrestricted text. If an agent creates a lead, for example, the business application might require a customer name, contact information, qualification status, assigned salesperson, and next action. That information can be validated before the system accepts it, making the interaction more dependable.

Once an agent can take action, controls become essential. **Authorization determines what the agent is allowed to access and do. Guardrails determine the conditions under which it should or should not do it.** A customer-service agent might be authorized to create a refund while a guardrail limits automatic refunds to a specified amount. A sales agent might be authorized to update CRM records but prevented from changing pricing or contractual conditions. These controls can reflect financial value, customer impact, information sensitivity, uncertainty, type of action, or business risk. Some actions can be performed automatically, others can be prepared by the agent and approved by a person, and situations outside established boundaries can be escalated. The objective is to give the agent enough authority to perform useful work without giving it more access or decision-making power than necessary.

Making Enterprise Agents Reliable

Real business environments are imperfect. Applications become unavailable, connections time out, information is incomplete, and actions sometimes fail. An enterprise agent must be designed for these situations rather than assuming every interaction will succeed. **Some failures can be handled through a retry.** If a system is temporarily unavailable, the agent may wait and try again. Other failures require a different action or escalation. The agent should not continue indefinitely or take increasingly risky actions simply because something failed.

Retries also introduce an important business risk: **an agent must not accidentally perform the same action twice**. Suppose an agent submits an order but does not receive confirmation because the connection fails. Simply repeating the request could create a second order. **Systems can use a mechanism known as idempotency to recognize that the action has already been requested and prevent an unintended duplicate**. The technical implementation can vary, but the business principle is straightforward: a retry should not result in a customer being charged twice, an order being created twice, or the same communication being sent repeatedly.

The organization must also be able to determine what its agents have done. **Auditability provides a record of important activities**: what information was accessed, which tools were used, what actions occurred, what approvals were required, and whether the task succeeded. This allows organizations to investigate problems, maintain accountability, evaluate performance, and improve the system over time. An agent that performs well in a demonstration is not necessarily ready to operate across a business. A demonstration might handle a handful of carefully selected situations; a production agent may need to handle hundreds or thousands of real interactions involving incomplete information, unusual requests, system failures, different users, security requirements, and unexpected conditions. Organizations therefore need monitoring, reliable system connections, controlled access, failure handling, and mechanisms for determining when human intervention is required.

Cost also matters. Agentic work can involve several model interactions, knowledge searches, and system actions before a task is completed. **Cost and usage controls** can govern which models an agent uses, how many steps it can take, how much AI usage a task can consume, and when continued activity should stop or be escalated. A simple principle applies: **a \$20 business activity should not require \$30 of AI resources to complete**. The operating cost of an agent should be considered against the value of the work it performs, just like any other business capability.

From Architecture to Business Capability

All these elements come together in the agent's **production architecture**. The architecture connects the AI model and its instructions with context, memory, enterprise knowledge, tools, APIs, business systems, identity and permissions, human oversight, monitoring, reliability mechanisms, and cost controls. The individual technologies may vary, but the business requirement remains the same: the agent must be able to understand the work, obtain the right information, perform authorized actions, handle problems, and operate in a way the organization can control and monitor.

FIGURE 6.1

Enterprise AI Agent Architecture

From business intent to real-world outcomes, securely and at scale.

PEOPLE
IDEAS
TECHNOLOGY
REAL IMPACT



The controls surrounding the architecture are not simply additional steps; they govern the entire operating environment. They dictate what data the agent can access, which tools it can use, what actions it can execute, and how it handles failures, monitors activity, or requests human intervention. For the user, however, the experience remains completely simple. A request or business event establishes the objective. From there, the agent automatically ingests the appropriate context, retrieves required knowledge, determines the next steps, executes the task using authorized tools, and validates the result before either finalizing the workflow or looping in a person.

The essential point is that building an enterprise AI agent is not about giving an AI model more freedom. It is about giving it the right information, tools, authority, and controls to perform useful work reliably. When those elements work together, an agent can move from an interesting demonstration to a practical business capability.

"The next logical question is how customers and employees will interact with these powerful capabilities. Increasingly, that interaction happens through the most intuitive interfaces people already use every day: text and voice. We explore this shift in the next chapter."

Voice, Chat and the New Customer Interface

For most people, the easiest way to interact with a business is not through software menus, forms, or complex applications. **It is simply to say or write what they need.** Artificial intelligence is making conversation an increasingly powerful interface between people and organizations. A customer or employee can communicate naturally through text or voice while AI interprets the request, retrieves relevant information, interacts with business systems, performs authorized actions, and involves a person when necessary.

This creates what we can call the **conversational enterprise**: an organization in which conversation becomes a way to access information, request services, initiate processes, and get work done. Businesses have used chatbots and automated telephone systems for years, but the important change is not simply that AI can hold a better conversation. **It is what AI can now do behind the conversation.**

From Conversation to Action

Text AI provides a simple example. A customer can type, “Where is my order?” instead of navigating through several pages of a website, while an employee can ask, “Which customers are waiting for a response?” instead of searching through a CRM. AI can interpret the request, retrieve the appropriate information, and respond naturally. When connected to enterprise systems and tools, it can go further by updating information, initiating a process, scheduling an activity, or completing another authorized action. The conversation is no longer limited to providing information; it can become the starting point for getting work done.

Realtime voice AI extends the same capability to spoken conversation. Instead of navigating a traditional telephone menu — “press 1 for sales, press 2 for service” — a customer can explain what they need in their own words. The AI can listen, interpret the request, respond, ask questions, retrieve information, and take appropriate actions while the conversation is taking place. **Modern speech-to-speech interaction** can make these exchanges faster and more natural by supporting interruptions, pauses, changes in direction, and the normal rhythm of conversation. Technologies such as **WebRTC** can provide the real time communication required to exchange voice and information with very little delay. Business leaders do not need to understand its technical mechanics; what matters is that the interaction can remain active while AI listens, responds, accesses information, uses tools, and reacts to what happens next.

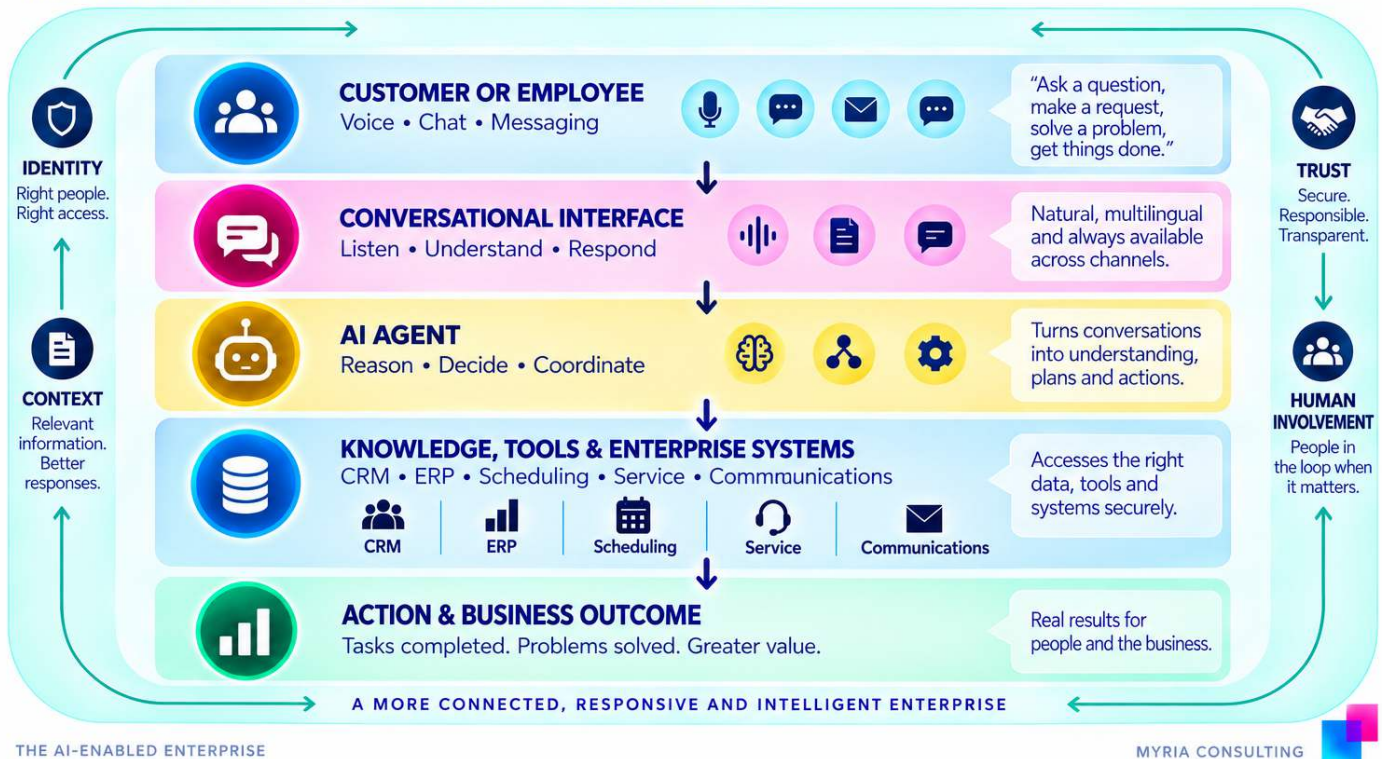
It is important, however, to **distinguish the conversation from the intelligence and actions behind it.** Voice, chat, and messaging are interfaces through which people interact with the organization. The AI agent behind that interface is what can interpret the request, determine what needs to happen, use enterprise knowledge and tools, and move the interaction toward an outcome. A natural-sounding voice alone does not create a valuable business capability. The true opportunity lies in linking conversation directly to the enterprise's ability to understand, decide, and act.

FIGURE 7.1

The Conversational Enterprise

Natural conversations. Real actions. Greater value.

PEOPLE
IDEAS
TECHNOLOGY
REAL IMPACT



The significance of this architecture becomes clear when conversation is connected directly to a business process. Instead of ending with an answer, the interaction can progress from **conversation** to **understanding**, from **understanding** to a **decision**, and from a **decision** to an **action** and **business outcome**.

From Customer Contact to Business Outcome

Consider a potential customer contacting a company. The agent can begin with **customer identification**, using information the person provides or information available through an authenticated interaction or approved business system. Depending on the situation, it might use an email address, telephone number, account information, or another permitted identifier to locate an existing record. If the person is new, the agent can collect the information required to create a prospect or lead. The level of verification should reflect the action being performed: providing general information may require little verification, while accessing an account or performing a sensitive transaction may require considerably more.

Once the customer and purpose of the interaction are understood, the agent can perform lead qualification by asking relevant questions, adapting subsequent questions to the answers, applying established qualification criteria, and determining the appropriate next step. A company receiving thousands of inquiries could use this capability to provide every prospect with an initial response while allowing salespeople to concentrate on opportunities where their expertise and personal involvement create the greatest value. If the prospect is qualified and ready to continue, the same conversation can move directly to appointment booking. The agent can check authorized calendars, offer available times, confirm the customer's choice, create the appointment, and provide confirmation.

Information gathered during the conversation can also be used for appropriate **CRM updates**. The agent can create or update the customer record, record qualification information, capture relevant notes, assign the opportunity, and establish the next action. With appropriate consent and business rules, the interaction can then continue through **SMS or other messaging channels**, where the agent might send appointment confirmations, requested information, reminders, or follow-up communications. What began as a conversation has now become part of a connected business process rather than an isolated interaction that an employee must manually process afterward.

The process does not always have to end with the AI. If the opportunity requires specialist attention, the agent can perform an **intelligent call transfer**. Rather than simply redirecting the call, it can identify why the person is contacting the organization, gather the necessary information, perform initial qualification or troubleshooting, determine the appropriate employee or team, and transfer the customer together with a concise summary of what has already happened. A salesperson or service representative can enter the conversation with useful context instead of asking the customer to begin again.

Human escalation serves a vital purpose: it establishes the boundaries of automation. An AI agent should immediately detect insufficient information, policy exceptions, or moments requiring critical human judgment. Far from a system failure, escalation represents optimal resource allocation. AI excels at handling predictable, transactional workflows, while humans step in when empathy, institutional authority, and complex problem-solving are required.

Creating a Connected and Trustworthy Experience

"Customers rarely view an organization through the lens of separate technology channels. They might start on a website, transition to chat, receive an SMS, and later speak with an employee—yet from their perspective, they are interacting with a single, unified business. **Omnichannel context** preserves vital history as the journey moves across text, messaging, voice, or human touchpoints. If a customer has already explained a problem in a chat window, they should never be forced to repeat themselves upon calling. Maintaining historical context, such as the reason for contact, verified data, and attempted actions, allows the next AI agent or employee to seamlessly resume the conversation exactly where it left off."

For employees, this continuity can be equally valuable. Instead of receiving an interaction with little context, a representative can receive a concise summary of the issue, the customer's identity when appropriately verified, relevant account information, what AI has already attempted, and why the situation was escalated. AI can therefore reduce repetitive information gathering for both customers and employees. Continuity does not mean every piece of information should follow a customer everywhere. Privacy, consent, identity, security, and business relevance still determine what can be retained and shared. The objective is **useful continuity with appropriate control**.

Trust extends beyond information management. A conversational AI experience should not be judged primarily by how human it sounds. A natural voice or well-written response is valuable only if the system is also useful, accurate, appropriately transparent, and capable of completing the work safely. The objective is not to make people believe they are speaking with a human; it is to make interacting with the organization easier and more effective. The system should protect sensitive information, verify identity when required, respect permissions, avoid commitments outside its authority, and provide a practical route to human assistance. When it does not understand, it should be able to ask for clarification, retrieve additional information, try an appropriate alternative, or involve a person rather than forcing the customer through an endless automated interaction.

Conversational AI should therefore be designed as part of the broader customer or employee experience rather than as an isolated chatbot or voice application. **The most important question is not simply, “Can AI have this conversation?”** It is:

“Can AI help this person accomplish what they came here to accomplish?”

Measuring What Matters

The same principle must guide how we measure conversational AI. While the volume of handled conversations or deflection rates offer useful operational data, neither metric proves that the customer experience or business outcomes have actually improved. Instead, performance must be directly tied to the interaction's purpose. In customer service, this means tracking resolution rates, first-contact resolution times, and customer satisfaction scores. In sales, the focus shifts to lead qualification quality, conversion rates, and revenue generation. Meanwhile, underlying technical health, such as response latency, tool execution failures, and abandoned sessions, serves a different purpose: pinpointing exactly where the architecture itself requires optimization.

Human escalation must be interpreted with caution. A lower handoff rate is not automatically better; an AI agent designed to resist escalation may create an illusion of autonomy while producing far worse outcomes. The superior metric is precision: ensuring the **AI effectively resolves what it can independently, while routing complex scenarios the moment human intervention adds higher value.** At its core, conversational AI exists to eliminate friction from the human experience. Customers should be empowered to express their needs naturally, transact effortlessly, and seamlessly access human support without barriers. Concurrently, employees must be equipped with rich context and actionable intelligence, shifting their role from simple message processing to high-value problem solving.

The conversational enterprise is not a mandate to replace human interaction; rather, it is a strategic shift to make conversation the ultimate gateway for people to leverage organizational capabilities and accelerate business outcomes.

As AI takes on increasingly complex workflows, however, a single agent may lack the full spectrum of knowledge, tools, or authority required. Achieving a strategic objective often demands diverse, specialized capabilities working in concert. **This operational evolution introduces the necessity of multi-agent systems and virtual AI teams.**

Multi-Agent Systems and the Virtual AI Team

A single AI agent can perform surprisingly complex work. It can gather information, reason about a situation, use tools, take authorized actions, and involve a person when necessary. But as the work becomes broader, more specialized, or more interconnected, asking one agent to understand every problem, access every system, follow every rule, and make every type of decision can become difficult to manage and control. In these situations, it may be more effective to divide the work among **specialized AI agents**, each designed for a particular responsibility.

This creates a **multi-agent system**: several AI agents working together toward a broader objective. **Think of it as a virtual AI team: specialized AI capabilities working together, each with a defined role, access to the information and tools it needs, and clear responsibilities within a larger business process.** One agent might specialize in customers, another in inventory, another in suppliers, and another in fulfillment. The goal is not to imitate a human organization or create as many agents as possible. It is to **divide complex work so specialization and coordination can produce better, safer, or faster outcomes.**

The comparison with a team has limits. AI agents are not employees and do not possess human accountability, experience, or organizational judgment. But the analogy helps reveal an important principle: once work is divided among specialized agents, the system must address many of the same operating questions that arise when coordinating any complex work. **Who should handle the task? What information do they need? How is work passed from one responsibility to another? What happens when conclusions conflict? Who validates important results? When should the work stop? And when should a person become involved?**

From One Agent to a Virtual AI Team

Consider a customer asking when an important delayed order will arrive. A single agent could potentially retrieve the order, check inventory, examine supplier information, review shipping options, evaluate alternatives, update the customer record, and prepare a response. In a relatively simple environment, that may be perfectly appropriate. As the organization becomes more complex, however, that same agent may require access to many systems, large amounts of information, numerous tools, and an increasingly complicated set of instructions and permissions. A virtual AI team can divide the problem differently. A **customer agent** can understand the request and retrieve the account and order, an **inventory agent** can determine whether the required products are available at another location, a **supplier agent** can examine expected deliveries and supplier commitments, and a **fulfillment agent** can evaluate shipping or transfer alternatives. Each agent can be given the information, tools, and authority appropriate to its role rather than giving one agent broad access to everything. Different agents may also use different AI models depending on the work: a routine information task may require relatively little reasoning, while evaluating an unusual operational exception may require greater capability.

"Specialization clarifies responsibilities and decision rights, but it introduces a critical requirement: **coordination**. When multiple agents contribute to a unified outcome, the system needs a clear way to determine which capabilities are required, where work should go, what information needs to follow it, and how the results should come together. In a centralized architecture, a master orchestrator receives the objective, delegates tasks to specialized agents, monitors progress, and synthesizes the results. Conversely, in a decentralized, event-driven model, agents respond directly to business events and pass work to one another dynamically without a single point of control. The enterprise architecture can therefore be centrally coordinated, collaborative, or a hybrid combination of both."

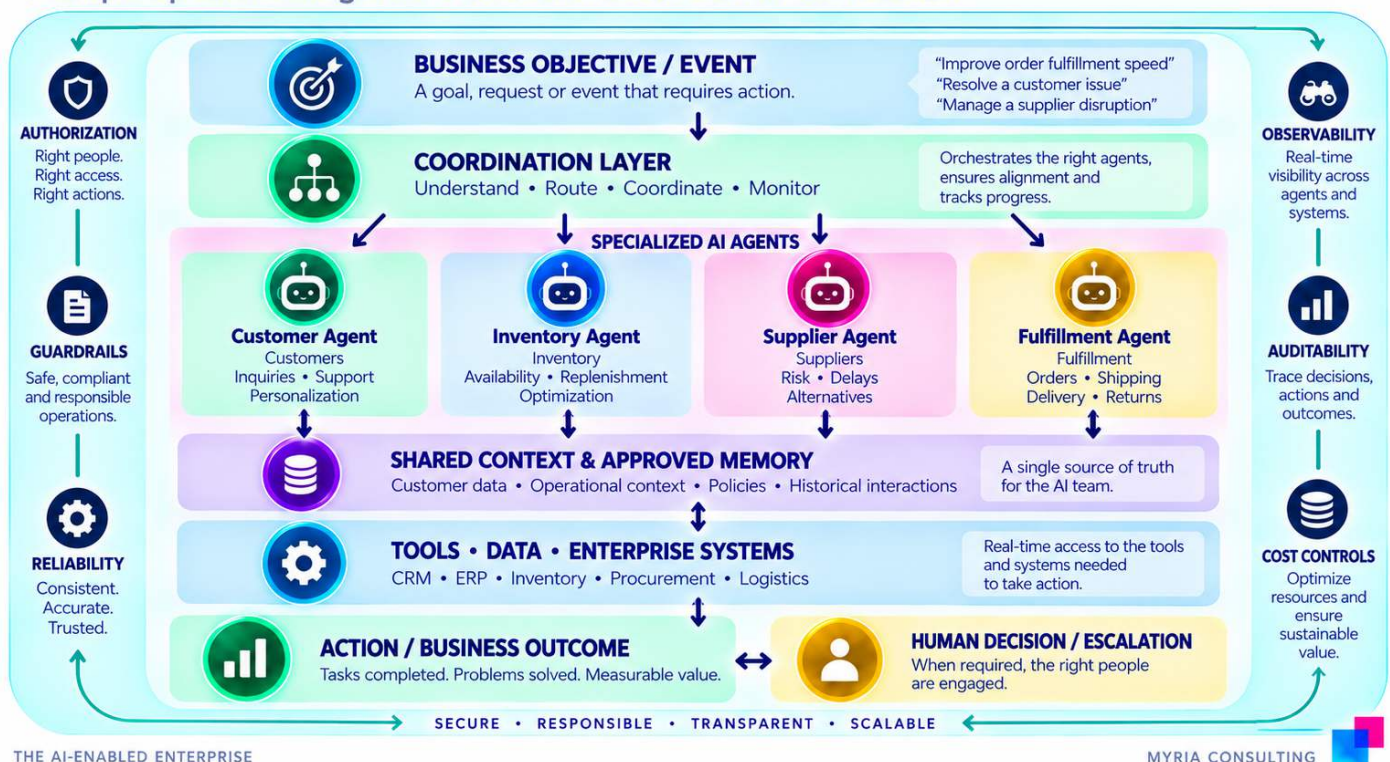
For instance, when handling a delayed order, the coordination layer might first route the problem to the order agent to establish exactly what happened. Once an inventory issue is identified, the coordination layer tasks the inventory and supplier agents with investigating alternatives. If another branch has stock while the supplier also has an incoming delivery, the fulfillment agent can evaluate which option is most likely to protect the customer commitment. **Agent routing** determines where the work should go, while **agent-to-agent handoffs** allow work or responsibility to move between specialized capabilities as the situation develops. For these handoffs to work, agents need enough **shared context** to understand the objective and what has already happened, including the order number, customer commitment, inventory status, previous actions, and results. Short-term memory can preserve information while the team works through the current situation, while longer-term memory can make approved information available for future interactions when appropriate. This does not mean every agent should have access to one unlimited pool of information. Context and memory should remain consistent with each agent's role, responsibilities, and permissions.

FIGURE 8.1

The Virtual AI Team

Multiple specialized agents. One coordinated team. Real results.

PEOPLE
IDEAS
TECHNOLOGY
REAL IMPACT



Coordinating Complex Work

Some work does not need to happen one step at a time. In the delayed-order example, the inventory agent can check stock at alternative locations while the supplier agent reviews incoming deliveries and the fulfillment agent examines transportation options. This parallel workflow significantly accelerates situational awareness, delivering multiple solutions much faster. Consequently, agents and business systems need reliable ways to exchange tasks, events, and outcomes—particularly when work occurs across different platforms or timelines. A message and event architecture supports this coordination. A business trigger such as an order delay, inventory shortage, payment failure, or high-value sales opportunity instantly activates the appropriate agent, while messages carry tasks and data asynchronously without requiring every system to remain active at the same moment. For the business reader, the principle is simple: the virtual AI team needs a dependable way to know that something happened, determine who should respond, and communicate what happens next. While parallel execution maximizes speed when activities can proceed independently, any sequence where one action depends on the result of another must still be strictly respected.

Coordination becomes even more important when specialized agents reach different conclusions. Suppose the inventory agent recommends transferring stock from another location because it would protect the customer commitment, while the fulfillment agent determines that expedited transportation would make the transfer unusually expensive. The system now has competing considerations: customer service, inventory availability, transportation cost, and perhaps the importance of the account. Conflict resolution determines how those alternatives are handled. Established business rules may resolve the situation, the coordination layer may compare the alternatives against defined priorities, or the decision may be escalated when the consequences exceed established limits. Some decisions can also benefit from an independent validation step, in which another AI capability checks whether required information is present, whether a proposed action follows established rules, or whether a recommendation is supported by the available information. These capabilities are sometimes described as validator or judge agents. Their purpose is not to make AI automatically correct, but to provide another layer of verification where it creates value. For consequential decisions, deterministic business controls or human approval may still be required.

Returning to the delayed order, the system might determine that transferring stock from another branch is the best solution. A validation agent could verify that the inventory is genuinely available, the proposed transfer meets established rules, and the customer commitment can realistically be achieved. If the additional cost is within the agent's authorized limit, the action might proceed. If it exceeds that limit, the recommendation can be presented to an employee for approval. The virtual AI team has investigated the problem, evaluated alternatives, and prepared the decision, while the organization retains human authority where it matters. The system must also know when to stop. Without clear stopping conditions, agents could continue searching, reconsidering alternatives, communicating with one another, or retrying failed actions without creating additional value. Work may stop because the objective has been achieved, a maximum number of attempts has been reached, necessary information cannot be obtained, a time or cost limit has been reached, or human judgment is required. Effective autonomy includes knowing when to continue, complete, or escalate.

Managing the Virtual AI Team

As more agents participate in business processes, organizations need visibility into how the work is being performed. Agent observability means being able to understand how an objective moved through the system: which agents became involved, what information and tools they used, how work was routed, where handoffs occurred, how long activities took, where failures happened, what actions were performed, and when a person became involved. But observability should ultimately serve business management, not merely technical monitoring. Leaders need to know whether the virtual AI team is accomplishing the outcomes for which it was created. In the delayed-order example, the important questions are not simply whether the agents successfully exchanged messages. Did the organization identify the problem sooner? How quickly was an alternative found? Was the customer commitment protected? What did the solution cost? How often did employees need to intervene? Were decisions consistent with business rules? The same principles apply across customer service, sales, procurement, finance, operations, and any other area where agents collaborate.

Designing a multi-agent enterprise architecture requires far more than simply connecting AI agents. Organizations must decide where specialization genuinely creates value, what responsibilities each agent should hold, and what information or tools it can access. Furthermore, they need clear blueprints for how work is assigned, how context will be shared, how conflicting conclusions are resolved, which results require validation, when people must become involved, and how the entire system will be monitored. **Ultimately, the number of agents is less important than the overall operating design. If a single agent can perform the work reliably, safely, and economically, one agent may be enough.** A multi-agent architecture becomes valuable when specialization, parallel workflows, distinct permissions, independent validation, or cross-functional coordination allow the enterprise to execute work far more effectively than a single agent could alone.

A well-designed multi-agent system therefore operates less like a collection of disconnected AI applications and more like a coordinated business capability. Within this system, specialized agents can investigate different parts of a problem, share appropriate context, work in parallel, evaluate alternatives, validate results, and take authorized actions—all while involving people when necessary and contributing to the same business objective. **The true value is not having more agents; it is enabling the organization to coordinate intelligence and action more effectively across increasingly complex operations.**

FROM AGENTS TO AUTONOMOUS OPERATIONS

The Autonomous Enterprise

AI agents can gather information, reason about situations, use business systems, take authorized actions, and coordinate with other specialized agents. These capabilities raise a larger question for organizations: **what happens when AI is no longer used only to complete individual tasks or respond to individual requests, but begins helping manage the continuous flow of decisions and exceptions that make up everyday operations?**

Every organization operates through thousands of decisions. Some are predictable and repetitive. Others arise because something has changed: a customer needs attention, a payment fails, demand shifts, an important sales opportunity is not followed up, a supplier misses a commitment, inventory becomes unavailable, or an approval is required. People continuously detect these situations, gather information, determine what happened, evaluate alternatives, make decisions, coordinate responses, and monitor the results. Much of the work inside an organization is therefore not simply executing processes. It is **responding to what happens while those processes are running.**

AI agents create the possibility of applying intelligence and action to this operating cycle. They can help detect situations requiring attention, gather the relevant context, evaluate alternatives, recommend or execute appropriate actions within established boundaries, monitor what happens, and involve people when their judgment or authority is required. **This is the foundation of autonomous operations.**

Autonomous operations are business operations in which AI can continuously observe what is happening, identify situations requiring attention, determine an appropriate response, and take authorized action within defined boundaries, while people retain control over objectives, rules, consequential decisions, and situations requiring human judgment.

From Automation to Autonomy

The distinction between **automation** and **autonomy** is important. Traditional automation is particularly effective when the appropriate response can be determined in advance: when X happens, do Y. Autonomy becomes relevant when the appropriate next action depends on the situation. AI can examine what has happened, gather additional information, consider available alternatives, and determine what should happen next within the authority it has been given. The difference is therefore not simply that AI performs more steps. **The system is participating in the decision about what should happen.**

This leads to an important way of thinking about autonomous operations: **the decision becomes a unit of automation.** Organizations have traditionally automated tasks and workflows. AI makes it increasingly possible to examine the decisions within those workflows: What decision needs to be made? What information is required? What alternatives are available? What rules and constraints apply? What are the consequences of the decision? Who makes it today? And under what conditions could AI support, recommend, prepare, or execute that decision?

Consider how broadly this applies. A sales process contains decisions about which opportunities require attention, which prospects appear qualified, what should happen next, and when a salesperson should become involved. Finance contains decisions around payments, invoices, discrepancies, credit, and approvals. Customer service involves decisions about priority, resolution, compensation, and escalation. A supply chain involves decisions about inventory, suppliers, orders, fulfillment, and changing demand. **The opportunity is not necessarily to automate an entire function. It is to identify recurring operational decisions where AI can help the organization respond more effectively.**

Managing Decisions and Exceptions

Some of the most critical decisions arise from **operational exceptions**: situations where actual events differ from what was expected, and someone must determine what to do. A service request remains unresolved past its deadline. A high-value opportunity misses a timely follow-up. A payment fails. A supplier changes a delivery commitment. Inventory levels threaten a customer order. In each case, the standard process alone is no longer enough; the standard workflow has broken down, and the situation demands immediate attention.

In many organizations, employees become the mechanism for handling these exceptions. They notice or receive an alert, search different systems, gather information, contact other people, compare alternatives, decide what to do, update systems, and follow up until the situation is resolved. This work can consume considerable time because the information, authority, and actions required to resolve one problem may be distributed across different functions and systems.

An autonomous operating capability can begin by **detecting what requires attention**. AI can monitor business events, transactions, commitments, thresholds, conversations, deadlines, and other operational signals to identify exceptions or opportunities. Detection can also become more proactive. A customer service system might recognize that an important request is approaching its service commitment. A sales capability might identify a valuable opportunity that has gone inactive. A finance agent might recognize a payment exception requiring investigation. In a supply chain, AI might detect that changing inventory levels, supplier information, and customer commitments are creating a risk of shortage or delayed delivery. The underlying capability is the same: **identify what matters early enough for the organization to respond effectively**.

Once a situation is detected, AI can **gather the context needed to understand it**. What happened? What was expected? Who or what is affected? What commitments have been made? What information is available? What actions have already been taken? Instead of requiring an employee to manually search several applications, documents, messages, or records, agents can retrieve and organize relevant information from the systems they are authorized to access. The objective is not simply to collect more data, but to assemble the information required to make the decision.

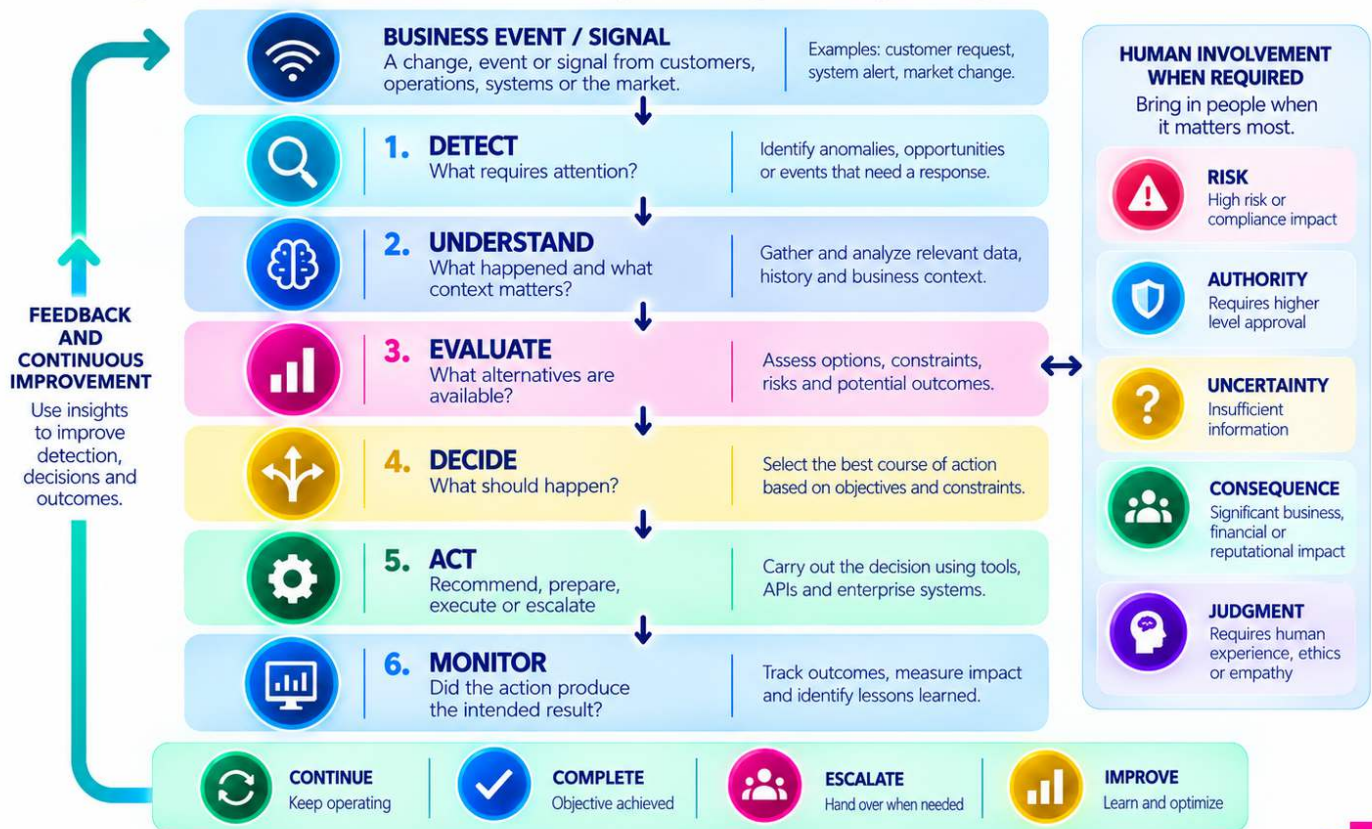
AI can then **evaluate alternatives** against business rules, priorities, costs, risks, and constraints. In sales, that might mean determining the appropriate follow-up for an opportunity. In customer service, it could mean identifying the best available resolution. In finance, it might mean determining how a payment exception should be handled. In a supply chain, it could mean considering available inventory, supplier commitments, or alternative fulfillment options. Depending on the decision and the authority granted to the AI, the system can recommend an action, prepare it for approval, execute it within defined limits, or escalate it to a person.

FIGURE 9.1

The Autonomous Operations Loop

From signals to outcomes — continuously learning and improving.

PEOPLE
IDEAS
TECHNOLOGY
REAL IMPACT



THE AI-ENABLED ENTERPRISE

PEOPLE + AI
A STRONGER TOMORROW

The true opportunity of autonomous operations is therefore not simply to automate more tasks. It is to **redesign how the organization detects what requires attention and understands the situation; makes recurring operational decisions and takes appropriate action; and monitors results while involving people where their judgment or authority creates the greatest value.**

Closing the Decision Loop

Taking an action is not the end of an autonomous process. The system also needs to **monitor the outcome**. If an AI capability follows up on an important sales opportunity, did the prospect respond? If it helps resolve a customer request, was the issue resolved? If it initiates a payment recovery action, was the payment received? If inventory is moved to protect a customer commitment, did the transfer occur and solve the problem without creating another one elsewhere? Monitoring closes the loop between **decision, action, and outcome**. When the intended result is not achieved, the system may need to gather new information, reconsider alternatives, try another permitted action, or involve a person. Autonomous operations therefore require the ability not merely to act, but to determine what should happen **after the action**.

This is where **escalation** becomes essential. An autonomous system should not be expected to resolve every situation. It needs to recognize when information is insufficient, uncertainty is too high, available actions fall outside its authority, attempts have failed, consequences exceed established limits, or human judgment is required. Escalation should not simply transfer a problem to a person with no explanation. Where possible, the system should provide the relevant context, what it has already investigated, the alternatives considered, the actions attempted, and why human involvement is required. AI can therefore reduce the work required to understand an exception even when the final decision remains human.

Over time, organizations can also **learn from outcomes**. This does not mean allowing agents to rewrite their own rules or expand their authority without oversight. It means using operational results to improve how the system works. Which recommendations were accepted or rejected? Which actions consistently resolved problems? Where did people intervene? Which exceptions occurred repeatedly? Where did the system lack the information required to make a good decision? Which actions produced unexpected consequences? These patterns can help improve processes, rules, instructions, thresholds, data, and the agents themselves.

This creates an important connection between autonomy and **continuous improvement**. If the same exception occurs repeatedly, the organization should not simply become better at resolving it. It should also ask why the exception continues to occur. Repeated customer escalations may reveal a service-process weakness. Frequent payment exceptions may expose a billing problem. Consistent human overrides may indicate that an agent's rules or authority need adjustment. Repeated inventory interventions may reveal a planning issue. **The objective is not simply to automate exceptions, but to learn from them and improve the operations that produce them.**

Human Decision Rights

Autonomy does not mean removing people from operations. It means deliberately deciding **which decisions AI can support, which it can execute, and which should remain with people**. Some decisions are repetitive, low-risk, reversible, and governed by clear rules. Others involve uncertainty, significant financial consequences, legal or regulatory obligations, safety, sensitive employee matters, major customer commitments, or reputational risk. The appropriate role of AI should depend on the nature and consequences of the decision, not simply on whether the technology is capable of performing it.

For one decision, AI may only gather information. For another, it may evaluate alternatives and recommend an action. It may prepare an action that requires human approval, or it may be authorized to decide and execute within clearly defined limits. When those limits are exceeded, the decision returns to a person. **Autonomy is therefore not an all-or-nothing condition. It can be assigned decision by decision according to value, risk, uncertainty, reversibility, and the need for human judgment.**

This transforms how human attention is deployed. By managing routine monitoring, data assembly, and bounded exceptions, AI frees employees from searching for information and coordinating predictable responses. Instead, human focus shifts to unusual scenarios, relationships, creativity, and decisions requiring judgment or accountability. The objective is not to eliminate people, but to deploy human and AI capabilities where each creates the greatest value.

The Autonomous Operating Model

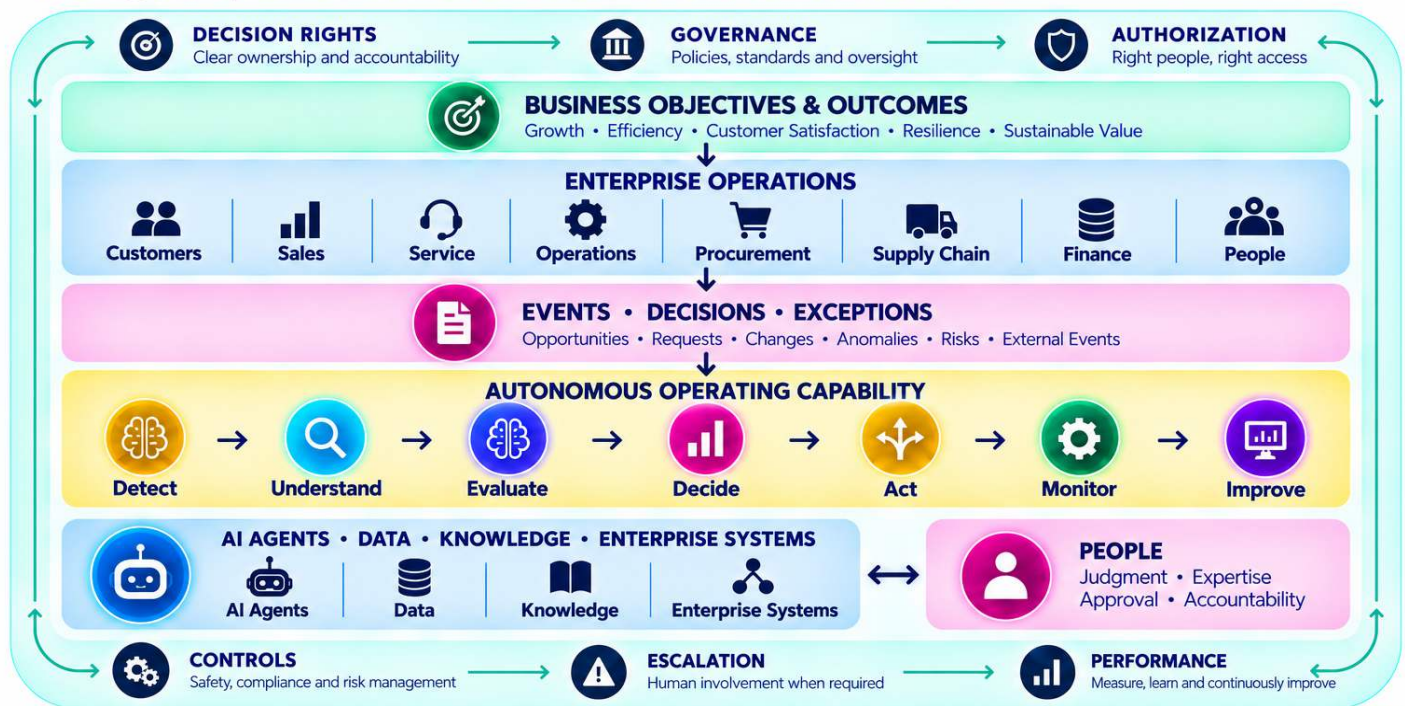
Expanding autonomous capabilities requires shifting from isolated agents to an Autonomous Operating Model. This framework defines how people, AI, processes, decisions, data, and systems interact. At its core are business objectives and operational exceptions. Within this model, AI monitors, evaluates, and acts; enterprise systems provide the data and tools; and people deliver judgment, expertise, authority, and accountability. Ultimately, decision rights, controls, escalation rules, performance measures, and governance determine the scope and execution of this autonomy.

FIGURE 9.2

The Autonomous Enterprise

Intelligent operations. Greater value. A more resilient future.

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The autonomous enterprise can therefore be understood as an organization that progressively develops the ability for **people and AI to manage operational decisions and exceptions together**, with the appropriate level of autonomy determined by the situation. These capabilities can operate across individual functions and, as they mature, become increasingly connected as decisions and actions in one part of the organization affect another.

The purpose of autonomy is not autonomy itself. A business gains little by making a process autonomous if the result is more expensive, less reliable, harder to control, or worse for customers and employees. The value comes from **earlier detection, faster and better-informed decisions, fewer unresolved exceptions, more consistent execution, reduced coordination effort, better use of information, and more time for people to focus on work where human capabilities matter most**.

The autonomous enterprise is therefore not defined by how much work AI performs. It is defined by how effectively people, AI, processes, data, and systems work together to detect what matters, understand what is happening, make appropriate decisions, take action, monitor outcomes, and continuously improve the performance of the organization.

The Autonomous Value Chain

Organizations are usually structured around functions. Sales manages customers and opportunities, finance manages financial activity, operations delivers products or services, human resources manages people processes, procurement works with suppliers, and customer service handles requests and problems. These structures provide specialization and accountability, but the value an organization creates rarely remains within a single function. **Value moves across the enterprise.**

A customer request may begin with sales but eventually involve operations, finance, service, and external partners. A supplier problem can affect operations, customer commitments, revenue, and cash flow. The organization may be divided into departments, but its customers, employees, partners, and other stakeholders experience the results of **the entire system**. This is why thinking in **value chains** becomes important. Instead of asking only how AI can improve an individual task, process, or function, organizations can examine how information, decisions, actions, commitments, and outcomes connect across the activities through which value is created. The opportunity is not simply to make each part of the organization more intelligent. It is to help the organization **understand and manage the relationships between those parts.**

Thinking in Value Chains

A value chain represents the connected activities through which an organization creates and delivers value. Its structure varies by organization: a manufacturer may connect suppliers, production, distribution, sales, and customers, while a service or public organization may connect people, expertise, programs, partners, processes, and those it serves.

One simple representation is:



The model reminds us that the enterprise does not operate in isolation. Information, resources, services, capital, commitments, risks, feedback, and exceptions flow continuously between the organization and its environment. The value chain is therefore not simply a one-directional sequence.

This matters because a business event is rarely important only to the function where it first appears. A sales commitment directly impacts capacity and delivery. A financial restriction alters what can be offered to a customer. A staffing shortage affects service levels, and a supplier delay jeopardizes production or customer commitments. **Understanding the value chain therefore means understanding these deep dependencies, not merely the sequence of activities.**

Understanding the Flows That Create Value

A value chain can also be understood through **the flows that connect it**. Customer demand, information, products, capital, decisions, and operational exceptions move continuously through different parts of the organization and its external relationships. These flows are deeply interconnected: increased demand generates additional revenue while simultaneously impacting capacity, staffing, purchasing, delivery commitments, and cash requirements. Similarly, a single pricing decision can influence demand, margins, and customer expectations. **The critical question is therefore not simply what happened in one function, but what it means for the rest of the value chain.**

AI can help make these relationships more visible by bringing together relevant information from different parts of the enterprise and helping people understand how an event or decision in one area may affect another. A changed supplier delivery date, for example, may initially appear to be a procurement issue. Its real importance depends on whether it affects production, available inventory, an important customer commitment, revenue, or another business outcome. In another organization, an unresolved customer issue, a staffing change, or a financial event may create a different chain of consequences.

Connecting Decisions Across the Enterprise

Value-chain relationships become particularly important when decisions are made. **A decision in one area can create opportunities, constraints, costs, or risks elsewhere.** Sales may want to make a special commitment to an important customer, but operations may need to determine whether it can be delivered and finance may need to understand the financial implications. What appears to be a decision belonging to one function may therefore depend on information and consequences across several parts of the enterprise.

AI can help connect these decisions by assembling the context that matters to the business outcome. A commercial decision might require information about the customer, demand, capacity, cost, margin, and delivery requirements. An operational decision might depend on current commitments, available resources, workforce capacity, financial constraints, and customer priorities. The objective is not to connect every decision to every piece of enterprise information. It is to **identify the relationships that matter and make the appropriate context available when it is needed.**

Individual AI capabilities can then contribute to a larger system when the appropriate connections exist between them. An event identified in one area can inform decisions elsewhere, bringing together the context required to understand the broader business outcome. A supplier issue, for example, might require operational, customer, and financial considerations to be viewed together, while an important sales opportunity might require commercial, capacity, and financial information before a commitment is made. **The architecture should follow the relationships that create value rather than connect systems or AI capabilities simply because they can be connected.**

FIGURE 10.1

The Connected Value Chain

People, partners and intelligent systems working together for greater value.

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A STRONGER, MORE CONNECTED TOMORROW

The important feature of this model is not the individual functions or the direction of the arrows. It is the interdependence among them.

Managing Value-Chain Exceptions

An exception becomes a value-chain issue when its consequences extend beyond the area where it first appears. A supplier delay may have little significance if sufficient capacity or inventory is available and no commitment is threatened, but the same delay may become important if it affects production, a major customer, revenue, or another critical outcome. An increase in demand may represent an opportunity for sales while creating pressure on capacity, staffing, service, or cash flow. **An enterprise exception should therefore be understood by its impact on the value chain, not simply by the department where it first appears.**

The important capability is to understand **how the consequences propagate through the enterprise**. Which activities, commitments, people, customers, partners, or financial outcomes could be affected? Which other decisions now become necessary? A change that appears minor in one area may become significant because of what it affects elsewhere.

This broader visibility can also reveal causes that are difficult to see from within a single function. What repeatedly appears as a customer-service problem may originate in fulfillment, product information, pricing, or another upstream activity. A recurring operational problem may originate in planning, staffing, supplier performance, or a policy elsewhere in the organization. Connecting signals across the value chain can help the organization address **causes across functional boundaries rather than repeatedly treating their downstream effects.**

Optimizing the Whole Enterprise

Connecting decisions across the value chain introduces another important challenge: the best decision for one part of the organization is not always the best decision for the organization as a whole. Sales may seek to maximize revenue, procurement may seek lower purchasing costs, operations may prioritize efficiency, customer service may seek faster resolution, finance may focus on margin or risk, and human resources may focus on workforce capacity. Each objective may be legitimate but optimizing one independently can create undesirable consequences elsewhere.

A low-cost supplier may reduce purchasing costs while increasing lead-time risk. Expedited fulfillment may protect an important customer relationship while reducing margin. Local optimization is not necessarily enterprise optimization.

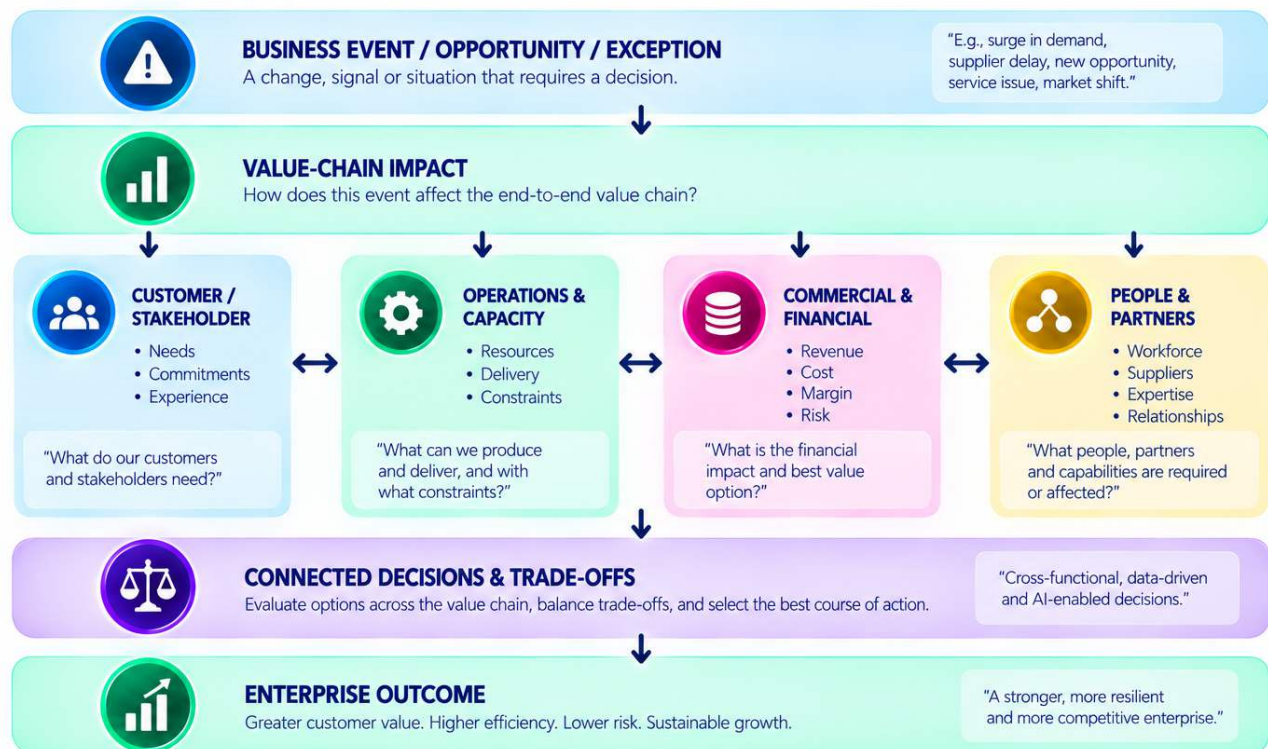
The larger opportunity is therefore to evaluate decisions in relation to enterprise outcomes and trade-offs. What creates the greatest overall value when customer impact, financial performance, capacity, risk, commitments, people, and other relevant considerations are taken into account? Leaders remain responsible for defining objectives and priorities. AI can help bring together the information and dependencies required to understand the consequences of different choices.

FIGURE 10.2

Connected Decisions Across the Value Chain

One event. Multiple perspectives. Better decisions. Greater value.

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PEOPLE + AI
A STRONGER TOMORROW



The question therefore changes from “What is the best decision for this department?” to “What decision creates the best overall outcome for the organization and the people it serves?” This shift is fundamental. Isolated AI applications can make individual activities faster or individual functions more efficient, while connected capabilities can help the organization understand dependencies, anticipate consequences, coordinate decisions across boundaries, and evaluate trade-offs that no single function sees completely. The autonomous value chain is therefore not defined by automating every activity from beginning to end. It is the progressive ability to connect the information, decisions, exceptions, actions, and outcomes through which the organization creates value, so that decisions can be made with a better understanding of their consequences across the enterprise. The greatest opportunity is not simply to make every part of the organization more intelligent but to make the organization more intelligent as a whole.

Designing the Autonomous Operating Model

Introducing AI agents into an organization does not by itself create autonomous operations. An agent may be able to analyze information, make recommendations, use business systems, or take actions, but those capabilities must operate within a larger organizational structure. Processes need to work, people need clear responsibilities, systems and data must be reliable, decisions need owners, authority must be defined, exceptions must be escalated appropriately, and performance must be measured.

The challenge is therefore not simply to determine what AI can do, but **how people, AI, processes, data, and systems should work together**. This requires an operating model that defines how work is performed, how decisions are made, where AI can act, where people remain involved, and how the organization maintains control as autonomy increases. An **autonomous operating model** defines how people, AI agents, processes, data, and enterprise systems work together to make decisions, execute work, manage exceptions, and deliver business outcomes within established boundaries. **The objective is not to automate as much as possible, but to design a better way of operating.**

Why Technology Alone Is Not Enough

Organizations have experienced this challenge with previous generations of technology. A new system may improve access to information without improving the process around it, while automation may accelerate one activity and leave delays elsewhere untouched. Technology can improve how work is performed, but it does not automatically redesign the organization around better outcomes. AI makes this challenge more consequential because it can participate in activities that previously depended heavily on people: interpreting information, evaluating situations, recommending decisions, coordinating work, and taking actions. As AI assumes more responsibility, questions about roles, authority, accountability, information, escalation, and performance become increasingly important.

An organization could therefore deploy sophisticated AI agents and still achieve disappointing results if the surrounding operating model is poorly designed. An agent may have the wrong information, insufficient access to systems, unclear instructions, excessive approval requirements, too much authority, or no effective way to handle situations outside its boundaries. **The quality of autonomous operations depends not only on the intelligence of the AI, but on the quality of the operating model around it.**

Designing Work Around Processes, Decisions and Exceptions

A practical starting point is to examine work through three connected elements: **processes, decisions, and exceptions**. Processes describe how work normally moves through the organization, decisions determine what should happen at important points, and exceptions arise when circumstances require a response that cannot simply follow the normal path. Traditional process design has often concentrated on activities: who performs each step, what system is used, what information is required, and what happens next. AI makes it useful to examine the **decisions embedded within those processes** as well. Which decisions occur repeatedly? What information is required? What rules or judgment are involved? What happens when information is incomplete or conditions fall outside the normal range? Which decisions require human authority, and which could be assisted or handled by AI?

Consider a customer request that requires special commercial condition. The process may involve reviewing the account, determining what can be offered, obtaining approval if necessary, communicating the response, and updating the appropriate systems. Within that process, however, are decisions about whether the customer qualifies, whether the request is financially acceptable, whether operational capacity exists, whether an exception is justified, and who has authority to approve it. Designing the operating model means determining how these decisions should be distributed among people, AI, and systems rather than simply automating the sequence of steps.

Some exceptions can be resolved through established rules, others can be evaluated by AI within defined boundaries, and some require human expertise, judgment, negotiation, or authority. The key question is therefore not simply **“Can this process be automated?”** but **“Which parts should remain human, which should be AI-assisted, and which can safely operate with greater autonomy?”**

Designing the Human + AI Operating Model

People, AI agents, systems, and data contribute different capabilities to the operating model.

People provide judgment, accountability, expertise, relationships, creativity, negotiation, and authority. AI agents can interpret situations, gather information, evaluate alternatives, coordinate activities, make recommendations, and take permitted actions. **Enterprise systems** provide the operational capabilities through which work is performed, while **data and knowledge** provide the context required to understand situations and make informed decisions.

These elements should be designed as parts of the same operating model rather than treated as separate technology and organizational questions. An agent supporting a customer decision, for example, may need access to customer history, policies, operational information, financial data, and approved business tools. It also needs to know what it can decide, what it can execute, what requires approval, and when the situation must be transferred to a person. When escalation occurs, the person should receive the relevant context rather than having to reconstruct the situation from the beginning.

The question is therefore not whether humans or AI should perform the work, but **how their different capabilities should be combined to produce the best business outcome**. AI may perform routine analysis while a person makes the decision, recommend and prepare an action for approval, or execute independently within defined limits and involve a person only when those limits are exceeded. The appropriate combination will vary by decision. Work that depends heavily on relationships, accountability, negotiation, creativity, or judgment may remain predominantly human, while repetitive, information-intensive, frequent work governed by clear boundaries may support greater AI involvement. **Designing the human + AI operating model means making these choices deliberately rather than allowing them to emerge from what the technology happens to make possible.**

Deciding How Much Authority AI Should Receive

Autonomy is not an all-or-nothing decision. Between fully manual work and autonomous operations are several useful levels at which AI can participate, giving organizations a practical way to decide how much authority AI should receive for a particular activity or decision.

The Myria AI Autonomy Maturity Framework defines six levels:

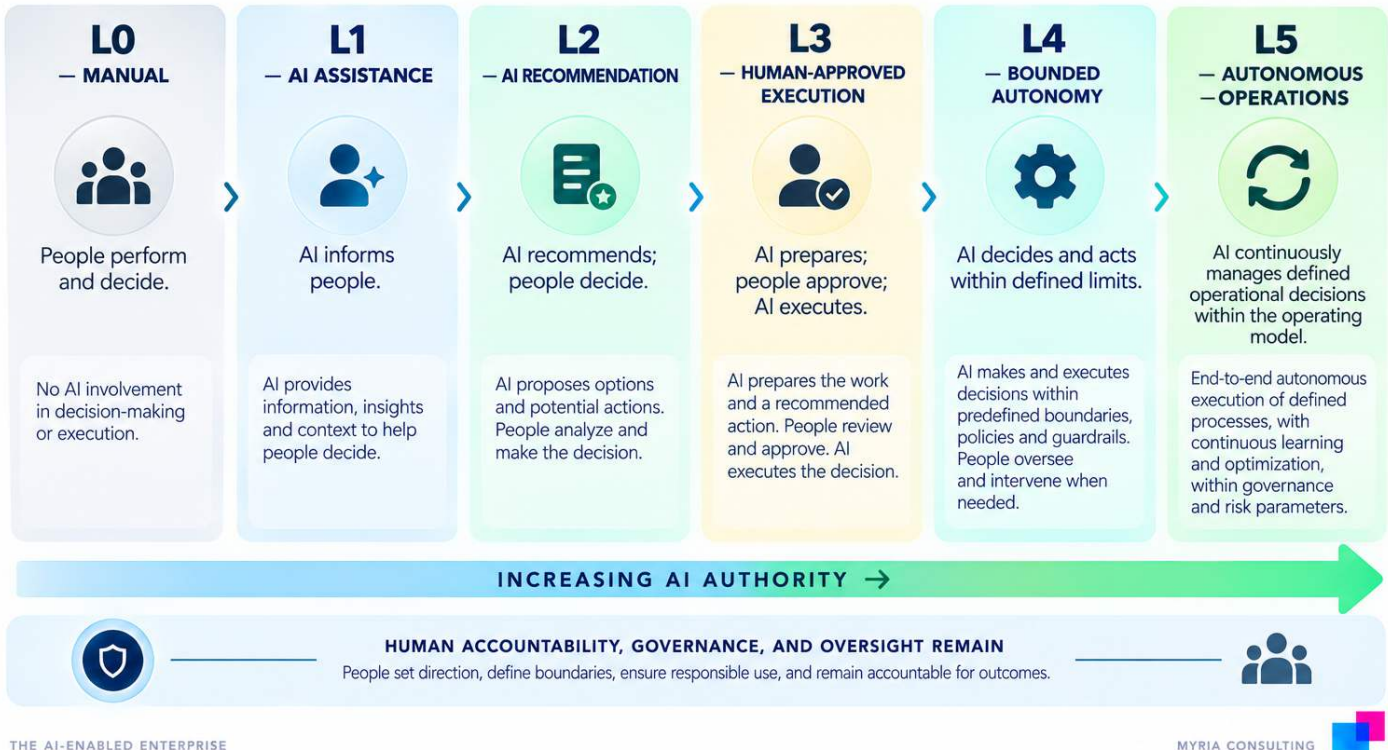
Level	Mode	Role of AI	Role of People
L0	 Manual	No meaningful AI involvement	People perform the work and make the decision
L1	 AI Assistance	Provides information, summaries, analysis, or support	People interpret the information and act
L2	 AI Recommendation	Evaluates the situation and recommends an action	People make the decision
L3	 Human-Approved Execution	Prepares the action and executes after approval	People approve, reject, or modify
L4	 Bounded Autonomy	Decides and acts independently within defined limits	People establish boundaries and handle escalations
L5	 Autonomous Operations	Continuously manages defined decisions and actions within the operating model	People establish objectives, oversight, and intervention mechanisms

FIGURE 11.1

Myria AI Autonomy Maturity Framework

From human-driven to AI-enabled to autonomous operations.

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Decision Rights, Governance and Escalation

Once an appropriate level of autonomy has been selected, it must be translated into clear **decision rights**. The operating model should define which decisions AI may make, which actions it may execute, what limits apply, which decisions require approval, and which situations must remain with people. These boundaries can be based on financial value, customer impact, risk, confidence, regulatory requirements, or the nature of the decision itself. An agent might resolve a routine service issue up to a defined value but require approval above that amount, or handle standard operational exceptions while escalating cases involving contractual changes, unusual risk, conflicting information, or important relationships.

Escalation is not a failure of autonomy, but part of its design. A well-designed system should recognize when a situation falls outside its authority, when information is insufficient, when confidence is too low, or when human judgment is required, and transfer the situation with the context already gathered, actions already taken, alternatives considered, and reason for escalation.

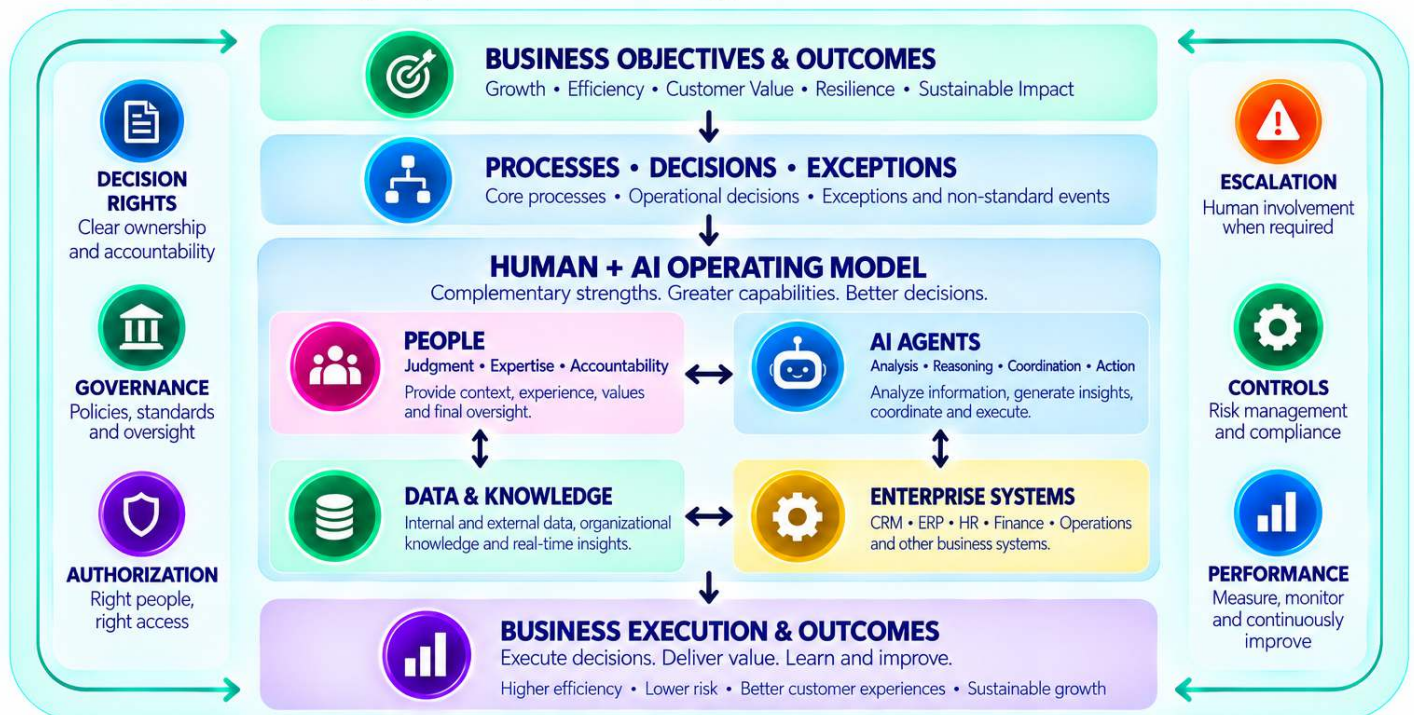
Governance provides the broader structure around these decision rights by establishing who defines and changes the boundaries, how performance is monitored, how decisions can be reviewed, and how the organization intervenes when necessary. As autonomy increases, effective governance becomes more important, not less.

FIGURE 11.2

The Human + AI Operating Model

People and AI working together to turn strategy into real outcomes.

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Managing and Improving the Operating Model

An autonomous operating model must ultimately be judged by its effect on the business. More tasks completed by AI, more automated decisions, or less human involvement do not necessarily mean better performance; autonomy creates value only when it improves meaningful business outcomes. Performance measures should therefore reflect the work being improved and may include cycle time, exception-resolution time, service levels, cost, quality, margin, customer experience, employee effort, or error rates. Measures such as escalation frequency, human overrides, failed actions, and approval times can also reveal how effectively the human + AI operating model is functioning

These measures provide a basis for continuous improvement. Frequent escalations may indicate that authority is too narrow or that human judgment is genuinely necessary; repeated overrides may reveal weak recommendations or missing context; slow approvals may expose unnecessary decision layers; and recurring exceptions may point to problems in the underlying process, policy, data, or system. The appropriate response may be to increase or reduce autonomy, redesign the process, improve the information available, or strengthen human involvement.

The operating model should therefore evolve as the organization learns. Decisions that begin at L1 or L2 may move toward L3 or L4 as reliability, data quality, performance, and organizational confidence improve, while others may remain at the same level because that provides the right balance of efficiency, control, and human judgment. Progress should be measured by better business performance, not by movement toward higher levels of autonomy.

Designing an autonomous operating model is ultimately about how work gets done when people and AI can both contribute to decisions and actions. Processes establish how workflows, data and systems provide information and capabilities, AI agents provide intelligence and execution, people provide judgment and accountability, and governance establishes the boundaries within which they operate. The organizations that use these capabilities effectively will not necessarily be those that give AI the most authority, but those that determine where AI creates value, where people create value, and how the two can work together to produce better outcomes.

BUILDING THE AI-ENABLED ENTERPRISE

AI Governance, Risk and Responsible Autonomy

As AI becomes more capable of participating in business operations, governance takes on a different importance. When AI primarily generates information, organizations must consider whether that information is accurate, appropriate, secure, and responsibly used. When AI agents can also make decisions, interact with enterprise systems, communicate with customers or employees, and take actions, the consequences of an error can extend beyond an incorrect response.

An agent might update the wrong record, communicate inaccurate information, initiate an inappropriate transaction, or expose information to someone who should not receive it. The issue is not that AI agents are inherently unsafe, but that **the consequences of AI output change when that output can become an action**. Responsible autonomy therefore requires organizations to determine what AI can access, what it can decide, what it can do, under which conditions it may act, when people must become involved, and how its activities can be monitored and controlled. **As AI moves from informing decisions to taking actions, governance must move from controlling outputs to controlling authority, actions, and consequences.**

Why Governance Changes When AI Can Act

AI risk can arise from **the information AI uses, the conclusions it reaches, the authority it receives, the actions it takes, and the consequences of those actions**. AI can misunderstand information, operate with incomplete context, reach an incorrect conclusion, or select an inappropriate action. Generative AI can also produce information that appears credible but is inaccurate, this is commonly referred to as **hallucination**.

For an AI assistant, an incorrect response may be reviewed and corrected before anything happens. For an agent with authority to act, the same error could influence a customer interaction, business record, financial decision, operational process, or another system. The practical governance question therefore becomes not simply **“Can the AI be wrong?”** but **“What happens if it is wrong?”** An incorrect internal summary and an unauthorized financial transaction do not create the same level of risk, just as a reversible action differs from a contractual commitment. Governance should therefore reflect **the risk and consequence of the decisions and actions AI is permitted to perform**, rather than applying the same controls to every use of AI.

When information is insufficient, conflicting, unusual, or outside established boundaries, the appropriate response may be to request additional information, restrict the available action, or involve a person. **Responsible autonomy requires AI systems to operate within uncertainty rather than behave as though uncertainty does not exist.**

Protecting Data, Systems and Access

AI agents can only perform useful enterprise work if they can access relevant information and systems, but that access creates responsibility. Access should therefore be designed around what an agent needs to perform its authorized responsibilities. **An AI agent should have no more access than it needs to perform its authorized role.** An agent that only needs to read information should not automatically be able to modify it, and access to one business capability should not provide unnecessary access to unrelated systems.

Identity and authorization are central to this model. The organization should be able to determine which agent is acting, on whose behalf, what it is authorized to access, and which actions it is permitted to perform. Data privacy requires similar discipline. When agents handle customer, employee, financial, confidential, or regulated information, organizations need to understand what information they can access, how it may be used and processed, where it may travel, how long it is retained, and what should be restricted. Security therefore becomes part of the agent's operating environment rather than a control added afterward. Authentication, permissions, data protection, system access, and tool restrictions should work together so that agents can perform their roles without receiving unnecessary authority. **Useful access and controlled access are not opposing objectives, but complementary requirements for reliable enterprise AI.**

Controlling Decisions and Actions

The autonomy levels introduced in the previous chapter provide a way to determine how much authority AI should receive. Governance translates that authority into **specific operating boundaries** by defining not only what an agent may do, but the conditions under which it may do it. An agent might independently resolve a routine issue below a defined threshold but require approval when financial value, customer impact, risk, uncertainty, sensitivity, or another condition exceeds that boundary. **Authority should be bounded not only by what an agent can do, but by the conditions under which it may do it.**

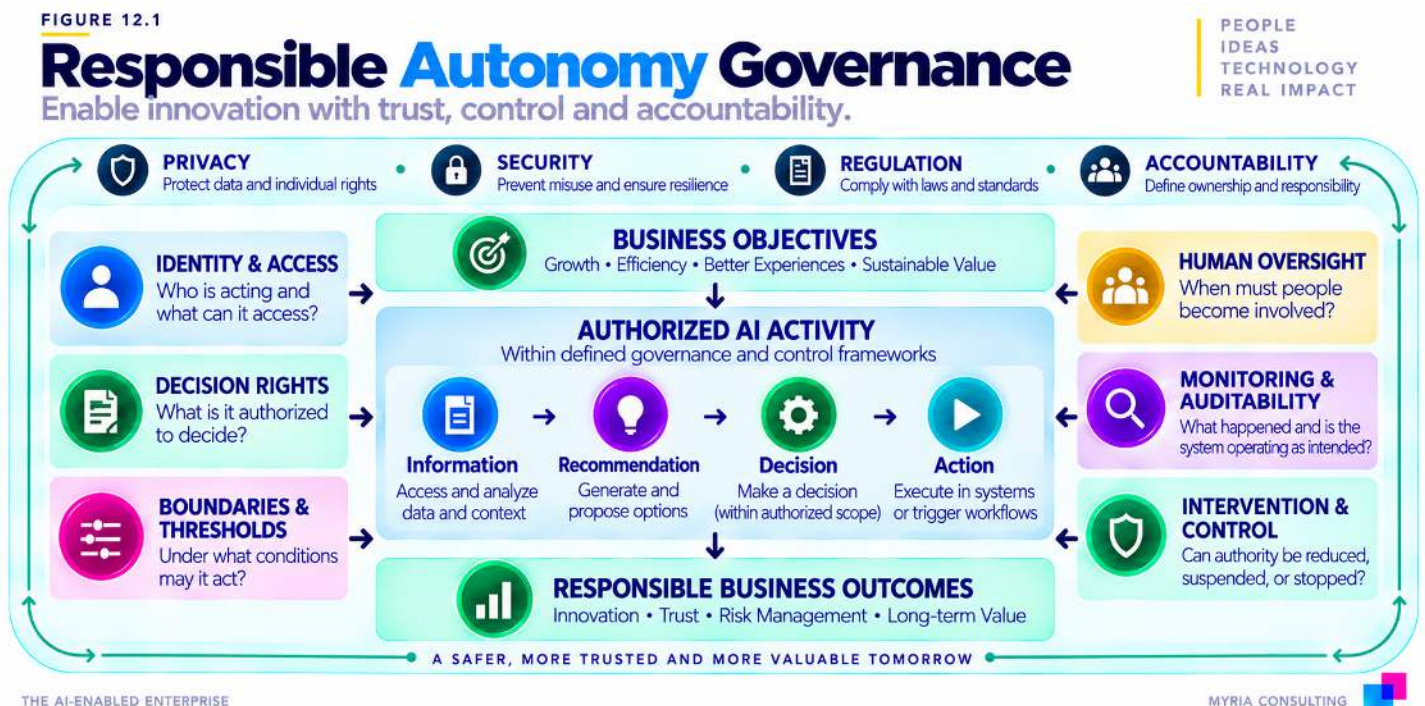
Approval thresholds and financial limits are practical ways to implement these boundaries, but not every threshold needs to be monetary. Customer or employee impact, confidence, reversibility, regulatory requirements, or the nature of the decision can also determine when human involvement is necessary. Human oversight should therefore be selective and purposeful. Requiring a person to approve every action can remove much of the value of autonomy, while removing people from every decision can create unnecessary risk. **Responsible autonomy places human involvement where the consequences justify it**, such as consequential actions, unusual situations, sensitive relationships, or circumstances outside established boundaries. The organization nevertheless remains accountable for defining what the system is intended to do, who owns the business outcome, what authority has been delegated, and how that authority is supervised.

Making Autonomous Operations Observable and Controllable

Organizations cannot responsibly delegate authority to systems they cannot observe. When an AI agent participates in business operations, it should be possible to reconstruct important aspects of what occurred: what triggered the activity, what information was used, what decision or recommendation was made, what action was taken, whether it was authorized, and what happened afterward. Audit trails provide this operational record and support review, investigation, compliance, and accountability.

Explainability complements auditability but does not necessarily mean exposing every technical detail of how an AI model operates. For business purposes, people often need to understand the relevant basis for an important recommendation or action: which information mattered, which rules or constraints applied, what action was selected, and why human approval or escalation was required. **The greater the consequence of a decision, the stronger the need for meaningful visibility into how it was reached.**

Monitoring extends that visibility from individual actions to the ongoing behavior of the system. Organizations may need to track failed or unusual actions, escalations, overrides, unexpected costs, and other signals that an agent is no longer operating as intended. Monitoring should connect technical behavior to business outcomes so that an agent that appears technically healthy but produces poor operational results does not go unnoticed. Organizations also need mechanisms for intervention, which may include pausing an agent, disabling a tool or action, revoking access, lowering its autonomy level, redirecting work to people, or shutting down the capability entirely. A kill switch is one form of intervention, but responsible control is broader than a single emergency mechanism. An autonomous system should never become more difficult to control simply because it has become more autonomous.



Building Responsible AI Governance

Individual controls are necessary, but responsible autonomy ultimately requires an organizational governance model. Organizations need to determine which AI capabilities can be introduced, who owns them, what authority they receive, what controls are required, and how their performance is governed over time. Governance should be proportionate to the use of AI: a low-risk internal assistant does not necessarily require the same controls as an agent capable of making financial commitments or taking actions affecting customers, employees, or regulated processes. **The level of governance should increase with the potential consequences of the AI activity.**

Regulatory considerations form part of this responsibility and vary by jurisdiction, industry, use case, information involved, and the consequences of the decision. Organizations may need to consider privacy and data-protection requirements, industry regulations, employment and consumer obligations, contractual requirements, and emerging AI-specific regulations. Responsibility must also remain clear: business leaders remain accountable for the outcomes produced by processes in which AI operates, while technology, security, privacy, legal, risk, compliance, and other functions contribute expertise and controls where appropriate. **Governance works best when accountability is clear rather than diffused across the organization or transferred implicitly to the technology.**

Responsible governance must also evolve through experience. Performance, incidents, regulatory developments, and changes in the business environment can reveal where boundaries or controls need to change. A capability that performs reliably may justify different operating limits over time, while another may require additional restrictions or human involvement. Governance therefore needs to support both **control and learning.**

Responsible AI governance is not about preventing AI from acting, but about creating the conditions under which AI can act safely, appropriately, and accountably. As organizations move toward greater autonomy, governance becomes part of the operating architecture itself: defining authority, protecting information and systems, establishing boundaries, maintaining visibility, enabling intervention, and ensuring that people remain accountable for the outcomes the organization chooses to pursue.

Finding the Right AI Opportunities

The growing capabilities of artificial intelligence can make it tempting to begin with the technology: identify what AI can do and then search for places to use it. That approach can produce interesting demonstrations without necessarily solving important business problems or creating meaningful value. A stronger approach begins with the organization. Where are customers waiting, employees spending unnecessary time, decisions taking too long, exceptions disrupting work, revenue opportunities being missed, or capacity being constrained? Once these situations are understood, AI can be evaluated as one possible way to address them. **The best AI opportunities usually begin with a business problem, decision, constraint, or opportunity, not with the technology.**

Start with Business Value, Not AI

AI initiatives should begin with an outcome the organization wants to improve, such as revenue, cost, productivity, capacity, quality, customer experience, employee effort, risk, or speed. Starting with the outcome creates a direct connection between the AI opportunity and the reason for investing in it. Asking “**Where can we use AI?**” encourages organizations to search for applications of a technology, while asking “**Where could better information, faster decisions, fewer exceptions, or more effective execution improve business performance?**” focuses attention on the work itself. AI can then be evaluated alongside alternatives such as process simplification, better data, system integration, conventional automation, policy changes, or eliminating unnecessary work. **The objective is not to find work for AI, but to find the best way to improve the business.**

The discovery sequence therefore becomes:



rather than:



Map Where Decisions and Exceptions Create Value

Once important business outcomes have been identified, map the value chain to locate where problems, delays, decisions, exceptions, and opportunities affect the creation and delivery of value. Within it, identify recurring decisions and exceptions that consume time, require judgment, delay outcomes, create unnecessary cost, or repeatedly demand coordination. Particular attention should be given to **repetitive cognitive work**: situations in which people repeatedly gather information, interpret what is happening, evaluate alternatives, determine what should happen next, and coordinate or execute a response. This might involve investigating a problem, determining whether a request qualifies, reviewing documents, comparing information across systems, prioritizing cases, or resolving routine exceptions.

The situations may differ, but the underlying pattern is often similar:



Frequency can make these opportunities more important than they initially appear. A decision requiring only a few minutes may consume substantial capacity when repeated thousands of times, while a small exception can create meaningful cost, delay, or customer impact when it occurs continuously. **Look for work where people repeatedly gather information, interpret a situation, make a decision, and determine what should happen next.**

Determine Whether the Opportunity Is Suitable for AI

An important business problem is not automatically a good AI opportunity. The organization must determine whether AI can realistically perform enough of the work to improve the outcome. **A good AI opportunity sits at the intersection of business value, technical feasibility, organizational readiness, and acceptable risk.**

The evaluation can be practical. Does the required information exist, and can it be accessed reliably? Can the necessary actions be performed through available systems? Is the work sufficiently repeatable and bounded? Is the process understood and ownership clear? Is the organization prepared to support the change? Is the risk acceptable for the proposed level of AI involvement?

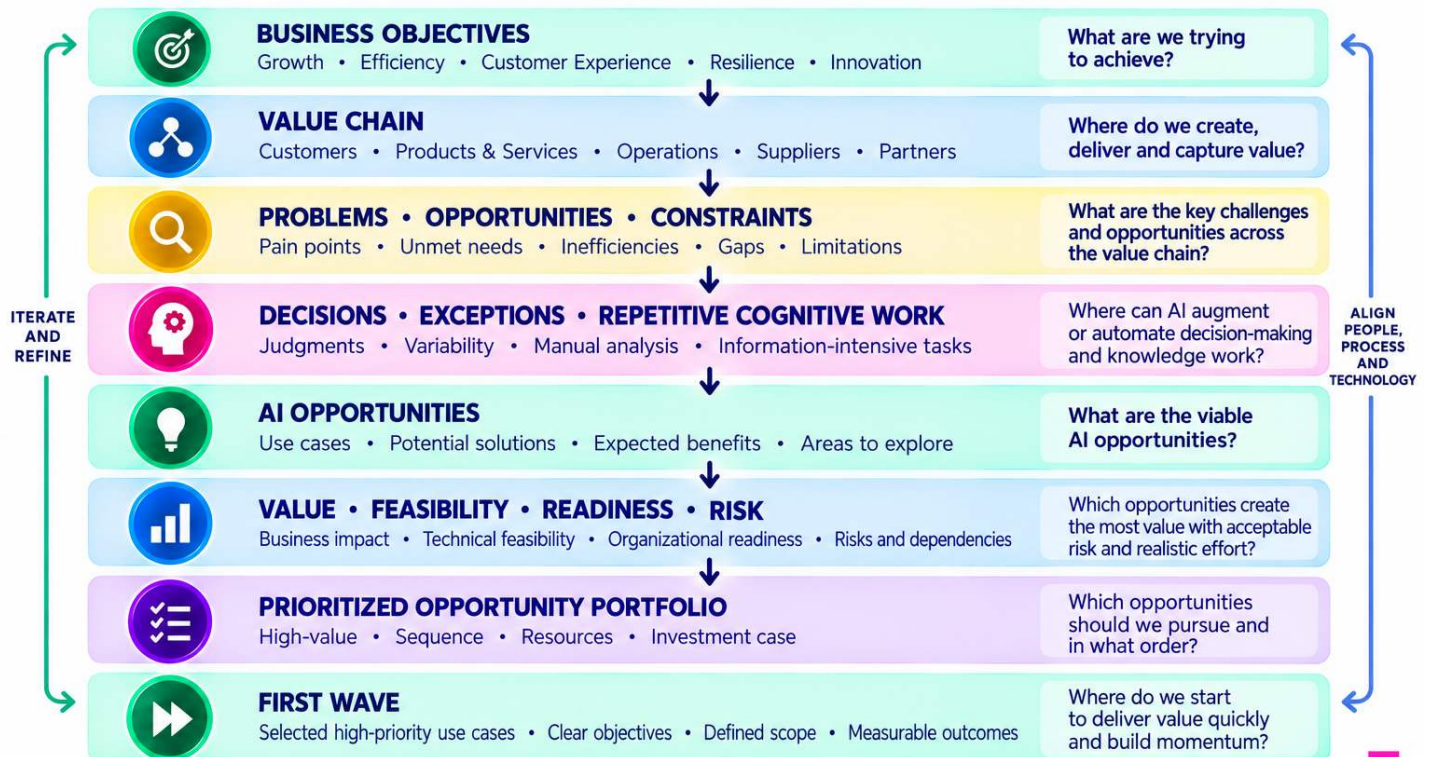
An initiative that depends on inaccessible information, unreliable records, or systems that cannot support the required actions may need foundational work first. Similarly, work involving high uncertainty, sensitive relationships, significant consequences, or extensive human judgment may be better suited to AI assistance or recommendation than autonomous execution. **An opportunity may offer substantial value but still be a poor first initiative if the organization is not ready to implement it reliably and responsibly, while something easy to automate may produce little meaningful value.**

FIGURE 13.1

Finding AI Opportunities

From strategy to a focused portfolio of high-value AI opportunities.

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Prioritize the AI Opportunity Portfolio

Organizations will often identify more AI opportunities than they can pursue at once, making the challenge not generating more ideas but determining which opportunities deserve investment. Business value should be considered broadly: an opportunity may increase revenue, reduce cost, release capacity, improve margins, accelerate work, strengthen customer experience, improve quality, or reduce risk. Wherever possible, these benefits should be translated into measurable outcomes so that expected value can later be compared with actual results.

A practical AI Opportunity Scorecard can help management teams compare opportunities consistently:

Criterion		What to Evaluate
	Business Impact	How significantly could the opportunity improve an important business outcome?
	Frequency	How often does the activity, decision, or exception occur?
	AI Potential	How much of the work could AI realistically assist, recommend, automate, or manage?
	Data Readiness	Is the required information available, accessible, and sufficiently reliable?
	Strategic Value	Does the opportunity support important priorities or create reusable capabilities?
	Risk	What are the potential consequences if the AI is incorrect or acts inappropriately?
	Complexity	How difficult will the opportunity be to implement and operate?

Each criterion can be scored using a simple scale, such as 1 to 5, but the purpose is not to manufacture a mathematically precise answer. The scorecard makes assumptions visible, creates a common basis for comparison, and supports management judgment.

The resulting opportunities can then be viewed as a portfolio based on value and readiness.

Figure 13.2 – AI Opportunity Portfolio



PRIORITIZE opportunities combine meaningful value with sufficient readiness. **PREPARE** opportunities offer potential value but require foundational work first. **SELECTIVE QUICK WINS** can provide measurable benefits, learning, or reusable capabilities, while **DEFER** opportunities generally do not justify immediate investment.

This portfolio perspective prevents organizations from selecting opportunities solely because they are technically interesting or easy to implement. **The easiest AI opportunity is not necessarily the best AI opportunity.** The better question is which combination can create meaningful value while strengthening the organization's ability to pursue larger opportunities over time.

Select the First Wave

The first wave turns prioritization into a manageable set of initiatives. The strongest candidates combine **meaningful business value, sufficient readiness, manageable risk, measurable outcomes, and reasonable implementation effort**. They do not necessarily need to be the organization's largest AI opportunities; they should demonstrate value while developing the experience required for broader adoption. A focused first wave can also create reusable capabilities.

Early initiatives may improve access to enterprise data, establish important system integrations, develop evaluation and governance practices, create reusable agent capabilities, and build experience in how people and AI work together. The organization can then apply what it learns to subsequent opportunities rather than treating every initiative as an isolated experiment.

The objective is not to build the longest list of AI use cases, but to identify the opportunities where AI can create meaningful value and where the organization is prepared to act.

From AI Experiment to Enterprise Transformation

Artificial intelligence is relatively easy to experiment with. A team can test a model, build a prototype, create an assistant, or demonstrate an agent capable of completing a useful task. The greater challenge is turning that experiment into a capability that works reliably within the organization and produces measurable business value. Many AI initiatives struggle at this transition because a pilot may demonstrate that AI can perform a task without proving that it can operate with real users, reliable data, enterprise systems, appropriate controls, clear ownership, and measurable outcomes. **A successful AI experiment proves that something is possible; a successful AI transformation proves that it creates value and can become part of how the organization operates.**

Move Beyond the Pilot

The path to enterprise transformation begins with the business problem. Instead of asking what an AI model or agent could demonstrate, the organization should define the outcome it wants to improve and understand the work that produces it. AI pilots often fail to scale when this connection is weak: the prototype may address a problem that is not important enough, depend on unavailable information or integrations, lack clear ownership, or work in controlled conditions but struggle with the uncertainty and exceptions of real operations. **The objective is not to move an AI prototype into production simply because it works, but to determine whether it can become a useful business capability.**

Discover and Design the Opportunity

Once an opportunity has been selected, discovery moves from identifying potential value to understanding what will be required to create it. The organization needs to understand the desired outcome, current work, people involved, relevant decisions and exceptions, information required, systems affected, constraints, risks, and baseline performance. **The purpose is not to document everything, but to understand enough of the operating environment to design the right solution.**

Architecture follows from this understanding. The organization can determine what AI capabilities are required, what knowledge and data they need, which tools and enterprise systems they must interact with, what people will continue to do, what authority AI should receive, and what controls are necessary. The solution may involve a single agent, multiple specialized agents, conventional automation, enterprise integrations, human approvals, or a combination of these elements.

Architecture should follow the business problem and operating model, not the other way around.

The progression from opportunity to design can therefore be understood as:



Prototype, Test and Prove Value

A prototype provides a practical way to test the most important assumptions before committing to broader implementation. It should be focused enough to build and learn quickly, but realistic enough to reveal whether the proposed capability can work under meaningful business conditions. An **AI Lab** can provide a controlled environment where business users, subject-matter experts, technology specialists, and other relevant participants test the capability against representative scenarios and increasingly realistic operating conditions.

Real users are essential because technical performance alone cannot determine whether a capability will work in practice. Validation should examine whether people can use it effectively, whether it operates reliably, whether the necessary information and systems are available, whether it handles expected exceptions, whether its boundaries are appropriate, and whether it improves the intended business outcome. A technically correct recommendation that employees cannot use effectively, or an agent that saves time in one activity while creating additional work elsewhere, does not represent successful implementation.

Governance and business value should be tested alongside the capability rather than addressed after the prototype succeeds. Decision rights, permissions, approval thresholds, human escalation, monitoring, auditability, and intervention mechanisms should operate under realistic conditions, while results should be compared with the baseline established during discovery. Depending on the objective, measures might include cycle time, employee effort, conversion, capacity, cost, errors, service levels, customer outcomes, or exception resolution.

Do not scale because the AI works; scale because the business capability works.

FIGURE 14.1

From AI Opportunity to Enterprise Transformation

A practical path from idea to lasting impact.

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TECHNOLOGY
REAL IMPACT



Scale Across the Enterprise

Once an AI capability has demonstrated that it can operate reliably and create measurable value, scaling can involve increasing transaction volume, extending the capability to additional locations or business units, supporting more decisions and exceptions, expanding authorized actions, or connecting it with other parts of the value chain. Success should continue to be measured as the scope expands because different environments may introduce new data, processes, regulations, customer expectations, or operating conditions.

The larger opportunity is to make what has been built reusable and increasingly connected. Data access, system integrations, knowledge sources, governance patterns, evaluation methods, monitoring capabilities, agent components, and operating practices developed for one initiative may support others. A capability designed for one business problem may also reveal related opportunities elsewhere, while independently developed agents and systems may eventually work together around shared outcomes. Instead of rebuilding the foundations for every project, the organization can progressively develop a common capability for implementing and operating AI across the enterprise. **The organization should scale capabilities, not simply applications.**

This requires organizational capability as well as technology. Leaders and employees need experience identifying opportunities, redesigning work, collaborating with AI, evaluating results, managing risk, and deciding where human judgment remains essential. Business, technology, data, security, risk, and other functions also need effective ways to work together. The organization becomes better at implementing AI because **its ability to design, govern, use, and improve AI becomes stronger.**

Build the AI-Enabled Enterprise

Enterprise transformation does not end when an AI capability is deployed. Business conditions change, technologies improve, employees learn, data becomes more useful, and new opportunities emerge. Continuous improvement therefore means examining whether AI continues to create value, where unnecessary effort or unresolved exceptions remain, where performance can improve, and whether the level of AI involvement remains appropriate. Experience may justify expanding AI authority in some situations and reducing it in others, while better information, integrations, or technology may make new opportunities feasible.

Over time, the organization's capabilities can evolve from:



This progression does not mean that every organization, process, or decision should move toward maximum autonomy. Different activities will require different combinations of people, AI, automation, and human judgment. An AI-enabled enterprise is therefore not defined by the number of AI systems it deploys or the amount of work it automates. **It is an organization that has developed the capability to identify where AI can create value, integrate it into how work gets done, govern it responsibly, measure the results, learn from experience, and continuously improve how people and AI work together.**

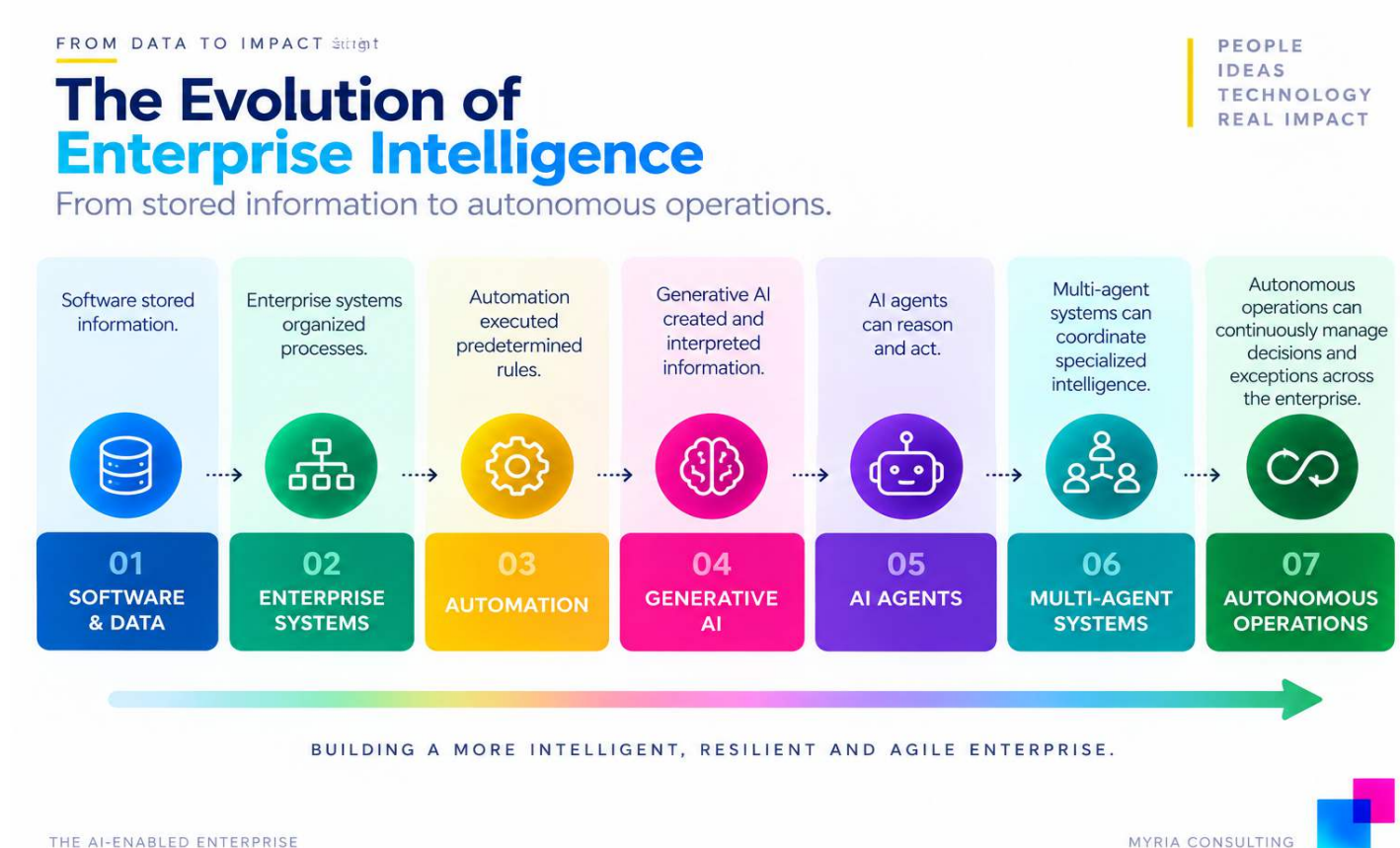
The goal is not to become an organization that uses AI everywhere. It is to become an organization that knows where and how to use AI to create value and has developed the capability to do so repeatedly, responsibly, and at scale.

CONCLUSION

The Enterprise That Thinks and Acts

Artificial intelligence represents another important evolution in how organizations use technology. For decades, technology has helped businesses store information, organize processes, connect systems, and automate work. AI extends that capability by allowing technology to participate more directly in **understanding, deciding, coordinating, and acting**.

The progression can be understood simply:



The AI-Enabled Enterprise

Each step expands what technology can contribute to the organization, but the opportunity is not simply to automate more work or give AI greater autonomy. People bring judgment, experience, creativity, empathy, relationships, and accountability, while AI can process information, identify patterns, perform repetitive cognitive work, interact with systems, and operate at speed and scale. **The greater opportunity is to determine how these capabilities can complement one another to improve how the organization operates and creates value.**

The AI-enabled enterprise is therefore not defined by how much work AI performs, but by the organization's ability to determine where people should lead, where AI should assist, where they should work together, and where AI can safely take greater responsibility within defined boundaries. **As these capabilities become connected across the enterprise, the organization becomes more responsive and better able to turn information into coordinated action.**

This changes the question leaders should be asking. The question is no longer simply:

“How can we use AI?”

It becomes:

“Which decisions, workflows and business outcomes should humans and AI manage together?”

The enterprise that thinks and acts is not one in which AI does everything. It is one in which people and AI work together to understand what matters, decide what should happen, take appropriate action, and continuously improve the outcomes they create.

THE FUTURE OF BUSINESS IS HUMAN + AI

The AI-Enabled Enterprise draws on our ongoing research and practical experience with artificial intelligence, combined with years of experience in organizational architecture and business transformation.

It offers a practical perspective on how organizations can move from understanding AI to applying it—connecting people, processes, information, and technology to improve how work gets done and create meaningful business value.

UNDERSTAND AI.
IDENTIFY THE OPPORTUNITIES.
CREATE BUSINESS VALUE.

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